When Does Regime Complexity Pay? Conditional Scoring Rules for Volatility Forecasts

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Abstract

Regime-switching volatility models are routinely evaluated by averaging loss functions unconditionally over the full out-of-sample period. We show that this practice can produce misleading model rankings when regime frequencies are imbalanced, as is typical in financial markets where calm periods substantially outnumber stress episodes. We develop a regime-conditional evaluation framework that preserves the strict consistency of proper scoring rules while enabling assessment of model performance within each regime. Our main theoretical results establish necessary and sufficient conditions for unconditional ranking reversals and identify when unconditional evaluation can be trusted. We extend the framework to the Model Confidence Set and to the Fissler-Ziegel loss for joint VaR/ES evaluation, and prove that regime-conditional tests using consistently estimated latent regimes are asymptotically valid. A Monte Carlo study confirms that ranking reversals arise in 33-48% of calibrated configurations. Applying the framework to a 21-asset volatility forecasting panel, we find that 59% of asset-horizon pairs exhibit ranking reversals: smooth transition models win unconditionally, but regime-conditional evaluation reveals that simpler models provide comparable or better forecasts during calm periods, while the regime-switching advantage is concentrated in stress episodes. These findings have direct implications for model selection in practice, particularly for risk management applications where forecast accuracy during stress episodes is of greatest economic importance.

1. Introduction

Regime-switching volatility models are evaluated almost exclusively by averaging a loss function over the entire out-of-sample period, yet the very premise of these models is that forecast dynamics are regime-dependent. This mismatch between conditional model structure and

unconditional evaluation is the starting point of this paper. When evaluation is unconditional, every observation receives equal weight regardless of the prevailing regime. Because calm periods substantially outnumber stress episodes in typical financial samples — a feature documented across equity, fixed-income, and currency markets (Hamilton 1989; Ang and Bekaert 2002) — average loss is dominated by performance during the tranquil state. A regime-switching model that delivers genuine improvements during stress, precisely the conditions it was designed for, may nonetheless be ranked below a simpler linear model that performs adequately during the calm periods that constitute the majority of the evaluation sample.

Standard practice follows naturally from the scoring-rule consistency results of Patton (2011), who proved that the QLIKE loss function correctly ranks volatility forecasts even when the true volatility is unobservable and only a noisy proxy is available. This is an important property, and our argument is not directed against it. Rather, we contend that the typical *application* of consistent scoring rules — unconditional averaging over the full sample — is incomplete when the data-generating process is regime-dependent. A scoring rule can be strictly consistent in the sense of Gneiting (2011) and Nolde and Ziegel (2017) and yet, when averaged unconditionally over a regime-imbalanced sample, produce model rankings that obscure or even reverse the ordering that would emerge within individual regimes. We formalize this observation and establish the precise conditions under which such ranking reversals occur.

The conditional evaluation of forecasts has received attention in the literature. Diebold and Mariano (1995) introduced the foundational framework for comparing predictive ac-

curacy, while Giacomini and White (2006) developed conditional predictive ability tests that condition on observable information sets. Hansen and Lunde (2005) and Hansen et al. (2011) proposed the Model Confidence Set for multiple model comparisons. Crucially, however, the Giacomini-White tests condition on *observable* information sets; no existing framework addresses the problem of conditioning on *latent* regime states — precisely the setting that arises with modern deep learning approaches to regime identification. Models such as Markov-switching GARCH (Hamilton 1989; Haas et al. 2004), smooth transition specifications (Teräsvirta 1994; Lundbergh and Teräsvirta 2002), and deep latent state-space approaches including variational recurrent neural networks (Krishnan et al. 2017; Rangapuram et al. 2018) all infer regime states from data rather than observing them directly, making the question of latent regime conditioning both practically relevant and theoretically nontrivial.

Our theoretical framework builds on the elicitability theory of Gneiting (2011), the consistent loss functions for volatility of Patton (2011), and the joint elicitability results of Fissler and Ziegel (2016) for Value-at-Risk and Expected Shortfall. We extend these foundations by developing a regime-conditional evaluation framework that preserves the desirable properties of proper scoring rules while enabling analysts to assess model performance separately within each regime. The contributions of this paper are fivefold. First, we prove that strictly consistent scoring functions remain strictly consistent under regime conditioning (Proposition 1), so that regime-conditional evaluation inherits the theoretical guarantees of unconditional proper scoring rules. Second, we derive necessary and sufficient conditions for unconditional ranking reversals (Theorem 1), characterizing exactly when two models can be correctly

ordered within each regime yet incorrectly ordered overall. Third, we identify sufficient conditions under which unconditional evaluation can be trusted to preserve regime-level rankings (Theorem 2), providing a practical diagnostic for analysts. Fourth, we extend the regime-conditional framework to the Model Confidence Set of Hansen et al. (2011) and to the Fissler-Ziegel loss for joint VaR/ES evaluation, showing that a regime-conditional Model Confidence Set can identify "stress specialists" — models that belong to the superior set during high-volatility regimes but not during calm periods, a distinction that is invisible to unconditional evaluation. Fifth, we prove that regime-conditional tests using consistently estimated latent regimes are asymptotically valid under regularity conditions specified in Section 2, ensuring that the framework applies to the growing class of models that employ neural network-based regime inference.

This final contribution connects our framework to the practical challenge of latent regime estimation. For models in the Markov-switching or deep state-space families, the regime indicator is not observed but inferred. We show that the regime-conditional evaluation remains valid provided the regime estimator satisfies a total-misclassification condition, and we discuss the connection to the generated-regressor literature and to the conditional predictive ability tests of Giacomini and White (2006). For risk management applications, unconditional backtesting of Value-at-Risk and Expected Shortfall systematically favors models that calibrate well during the tranquil periods that dominate the sample, potentially at the expense of models that provide better tail coverage during the stress episodes when risk management matters most.

Our empirical application leverages the volatility forecasting results of Lim (2026a), which

evaluated six competing models — HAR, STR-HAR with observable and latent transitions, GARCH(1,1), EGARCH(1,1), and Markov-switching GARCH — across a panel of 21 assets spanning equity indices, sector ETFs, government rates, major currency pairs, and commodities, at forecast horizons of 1, 5, and 22 days. Applying our regime-conditional evaluation framework to this dataset, we find that ranking reversals occur for 59 percent of asset-horizon pairs. In the majority of reversals, smooth transition models win unconditionally but lose in the calm regime to simpler specifications, while their advantage is concentrated in stress episodes. These findings demonstrate that the choice of evaluation framework can influence which model is deployed, and that unconditional evaluation may lead to suboptimal selections in precisely the conditions when forecast accuracy matters most.

The remainder of this paper is organized as follows. Section 2 develops the theoretical framework, including the regime-conditional proper scoring rules, the ranking reversal theorems, and extensions to the Model Confidence Set and risk evaluation. Section 3 presents a Monte Carlo simulation study that quantifies the frequency and magnitude of ranking reversals under controlled conditions; the simulations confirm that reversals arise regularly in calibrated settings where the regime-switching model is the true data-generating process. Section 4 applies the framework to our 21-asset empirical panel. Section 5 discusses the practical implications and limitations, and Section 6 concludes.

2. Theoretical Framework

This section develops the formal theory of regime-conditional proper scoring rules for volatility forecasts. We begin with a review of elicitability and proper scoring rules in Section 2.1, drawing on the foundational results of Gneiting (2011) and Patton (2011). Section 2.2 introduces conditional elicitability under regime-switching and establishes that strict consistency is preserved under regime conditioning. Section 2.3 derives conditions under which unconditional scoring rules produce ranking reversals—cases where the unconditional winner is the conditional loser in the dominant regime. Section 2.4 establishes sufficient conditions for unconditional and conditional rankings to agree. Section 2.5 extends the Model Confidence Set of Hansen et al. (2011) to regime-conditional settings. Section 2.6 treats the joint elicitability of Value-at-Risk and Expected Shortfall under regime conditioning, building on Fissler and Ziegel (2016). Section 2.7 connects the framework to the conditional predictive ability tests of Giacomini and White (2006), addressing the additional complications that arise when the conditioning event involves a latent state.

Notation. Throughout, n denotes the sample size and $T : \mathcal{F} \to \mathbb{R}$ denotes a statistical functional. We reserve T exclusively for functionals to avoid ambiguity with the sample size.

2.1 Elicitability and Proper Scoring Rules

We begin by establishing notation and reviewing the core results from the elicitability literature. Let Y denote a real-valued random variable with distribution $F \in \mathcal{F}$, where $\mathcal{F}$ is a convex class of probability distributions on $\mathbb{R}$. A statistical functional $T : \mathcal{F} \to \mathbb{R}$ maps dis-

tributions to real numbers; familiar examples include the mean $T(F) = \mathbb{E}_F[Y]$, the variance $T(F) = \mathrm{Var}_F(Y)$, and quantiles $T(F) = F^{-1}(\alpha)$.

A scoring function $S : \mathbb{R} \times \mathbb{R} \to [0, \infty)$ maps a point forecast $\hat{y}$ and a realized value y to a non-negative loss. The scoring function is said to be *consistent* for T if the expected score is minimized at the true functional value, and *strictly consistent* if the minimizer is unique. These properties are formalized in the following definition, which follows Gneiting (2011).

Definition 1 (Consistency and strict consistency). A scoring function $S : \mathbb{R} \times \mathbb{R} \to [0, \infty)$ is *consistent* for the functional $T : \mathcal{F} \to \mathbb{R}$ relative to $\mathcal{F}$ if, for all $F \in \mathcal{F}$ and all $x \in \mathbb{R}$,

$$\mathbb{E}_F[S(T(F), Y)] \leq \mathbb{E}_F[S(x, Y)].$$

The scoring function is *strictly consistent* for T if equality holds only when $x = T(F)$.

The concept of elicitability, introduced formally by Lambert et al. (2008) and developed extensively by Gneiting (2011), characterizes which functionals admit strictly consistent scoring functions.

Definition 2 (Elicitability). A functional $T : \mathcal{F} \to \mathbb{R}$ is *elicitable* relative to $\mathcal{F}$ if there exists a scoring function S that is strictly consistent for T relative to $\mathcal{F}$.

Gneiting (2011) showed that the mean, quantiles, and expectiles are elicitable, while the variance is elicitable as a function of the first two moments. Critically for volatility forecasting, Gneiting (2011) established that the variance functional $T(F) = \mathrm{Var}_F(Y) = \mathbb{E}_F[Y^2] - (\mathbb{E}_F[Y])^2$ is not directly elicitable as a standalone functional but becomes elicitable when the mean is also specified. Patton (2011) resolved this for the volatility forecasting set-

ting by showing that the conditional expectation of a volatility proxy is elicitable, and that certain loss functions maintain consistent model rankings even when the true volatility is unobservable.

The loss function that is central to our analysis is the QLIKE loss, which belongs to the Bregman divergence family. For a volatility forecast $\hat{\sigma}^2$ and a realized proxy σ^2, the QLIKE loss is defined as

$$S_{\text{QLIKE}}(\hat{\sigma}^2, \sigma^2) = \frac{\sigma^2}{\hat{\sigma}^2} - \log \frac{\sigma^2}{\hat{\sigma}^2} - 1.$$

Patton (2011) proved that S_{QLIKE} is strictly consistent for the conditional variance under mild regularity conditions. Moreover, and crucially for empirical work, Patton (2011) showed that the ranking of forecasts by expected QLIKE loss is *robust to noise in the volatility proxy*: if we replace the latent true variance with a conditionally unbiased proxy (such as realized variance computed from high-frequency data), the ranking of forecasts is preserved. This robustness property makes QLIKE the workhorse loss function for volatility forecast comparison. We note that proxy robustness requires conditional unbiasedness of the proxy with respect to the information set on which the forecasts are based. When we condition on regime state, this requires that the proxy remain conditionally unbiased within each regime, a condition that is satisfied when the proxy is constructed from intraday data that are available regardless of the regime state.

More broadly, Patton (2011) established that any member of the homogeneous Bregman

divergence family,

$$S_b(\hat{\sigma}^2, \sigma^2) = \begin{cases} \frac{1}{(b+1)(b+2)} \left[\sigma^{2(b+2)} - \hat{\sigma}^{2(b+2)}\right] - \frac{1}{b+1}\hat{\sigma}^{2(b+1)}(\sigma^2 - \hat{\sigma}^2) & b \neq -1, -2 \\ \frac{\sigma^2}{\hat{\sigma}^2} - \log \frac{\sigma^2}{\hat{\sigma}^2} - 1 & b = -2 \\ \sigma^2 \log \frac{\sigma^2}{\hat{\sigma}^2} + \hat{\sigma}^2 - \sigma^2 & b = -1 \end{cases}$$

yields a scoring function that is robust to proxy noise, with QLIKE corresponding to $b = -2$ and squared error to $b = 0$.

The standard approach to forecast evaluation applies these scoring rules unconditionally. Given a sample of n forecast-realization pairs $\{(\hat{\sigma}_t^2, \sigma_t^2)\}_{t=1}^n$, the unconditional expected loss is estimated by the sample average,

$$\bar{S} = \frac{1}{n} \sum_{t=1}^{n} S(\hat{\sigma}_t^2, \sigma_t^2),$$

and models are ranked by their average loss. The Diebold-Mariano test (Diebold and Mariano 1995) provides a formal comparison by testing whether the mean loss differential between two models is zero. This unconditional evaluation is the standard in the volatility forecasting literature, as exemplified by Hansen and Lunde (2005)'s comparison of 330 volatility models and the widespread use of QLIKE as a ranking criterion.

The implicit assumption underlying unconditional evaluation is that a model that minimizes average loss across all periods is the preferred model for all purposes. As we show in the sections that follow, this assumption can fail when forecasting models are designed to perform differently across regime states.

2.2 Conditional Elicitability Under Regime-Switching

We now extend the elicitability framework to regime-conditional settings. Let $G_t \in \mathcal{G}$ denote a regime state at time t, where $\mathcal{G}$ is a finite or compact set of possible regime values. In the discrete case, $\mathcal{G} = \{1, 2, \dots, K\}$ represents K distinct regimes (e.g., low and high volatility states). In the continuous case, $\mathcal{G} = [0, 1]$ represents a smooth regime index such as the transition function value in an STR model. For concreteness, we develop the theory for the discrete case and note where the continuous extension requires additional care.

Let $F_{Y|G=g}$ denote the conditional distribution of Y given regime state $G = g$, and let $T_g \equiv T(F_{Y|G=g})$ denote the functional of interest evaluated under this conditional distribution. For volatility forecasting, $T_g = \mathbb{E}[\sigma^2_{t+h}|G_t = g]$ is the conditional variance given regime state g.

The regime-conditional expected loss for a forecast $\hat{y}$ under regime g is

$$L(g; \hat{y}) = \mathbb{E}[S(\hat{y}, Y) \mid G = g].$$

The key question is whether the scoring function S, which is strictly consistent for T under the unconditional distribution F, remains strictly consistent for the conditional functional T_g under the conditional distribution $F_{Y|G=g}$. The following proposition establishes that strict consistency is indeed preserved under conditioning.

Proposition 1 (Conditional consistency of proper scoring rules). Let $S : \mathbb{R} \times \mathbb{R} \to [0, \infty)$ be a strictly consistent scoring function for the functional $T : \mathcal{F} \to \mathbb{R}$ relative to a convex class $\mathcal{F}$. Let G be a random variable taking values in $\mathcal{G}$ such that, for each $g \in \mathcal{G}$, the conditional

distribution $F_{Y|G=g}$ belongs to $\mathcal{F}$, and $\mathbb{E}[S(x, Y) \mid G = g]$ exists and is finite for all x in a neighborhood of $T(F_{Y|G=g})$. Then for each fixed $g \in \mathcal{G}$:

(i) S is strictly consistent for $T_g = T(F_{Y|G=g})$ under the conditional distribution $F_{Y|G=g}$, that is,

$$\mathbb{E}[S(T_g, Y) \mid G = g] < \mathbb{E}[S(x, Y) \mid G = g] \quad \text{for all } x \neq T_g.$$

(ii) For Bregman divergences, strict consistency implies that $L(g; x) = \mathbb{E}[S(x, Y) \mid G = g]$ is a strictly convex function of x at the minimizer T_g, so that $|\hat{y}_A - T_g| < |\hat{y}_B - T_g|$ in Euclidean distance implies $L(g; \hat{y}_A) < L(g; \hat{y}_B)$ for predictions sufficiently close to T_g (Banerjee et al. 2005).

Proof. The proof follows from the definition of strict consistency applied to the conditional distribution. Fix $g \in \mathcal{G}$ and let $F_g = F_{Y|G=g}$. By assumption, $F_g \in \mathcal{F}$. Since S is strictly consistent for T relative to $\mathcal{F}$, we have, for all $x \neq T(F_g)$,

$$\mathbb{E}_{F_g}[S(T(F_g), Y)] < \mathbb{E}_{F_g}[S(x, Y)].$$

But $\mathbb{E}_{F_g}[S(x, Y)] = \mathbb{E}[S(x, Y) \mid G = g]$ by the definition of the conditional distribution. Therefore,

$$\mathbb{E}[S(T_g, Y) \mid G = g] < \mathbb{E}[S(x, Y) \mid G = g] \quad \text{for all } x \neq T_g,$$

which establishes part (i). For part (ii), Banerjee et al. (2005) showed that every Bregman divergence generates a strictly consistent scoring function whose expected loss is a strictly convex function of the prediction. Specifically, for a Bregman divergence $S_\phi(x, y) = \phi(y) -$

$\phi(x) - \phi'(x)(y - x)$ generated by a strictly convex function ϕ, the conditional expected loss $L(g; x) = \mathbb{E}[S_\phi(x, Y) \mid G = g]$ inherits strict convexity from ϕ. By strict convexity of $L(g; \cdot)$ at its minimizer T_g, for any predictions $\hat{y}_A, \hat{y}_B$ in a sufficiently small neighborhood of T_g, the ordering $|\hat{y}_A - T_g| < |\hat{y}_B - T_g|$ implies $L(g; \hat{y}_A) < L(g; \hat{y}_B)$. This local monotonicity in Euclidean distance follows from the positive definiteness of the Hessian of $L(g; \cdot)$ at T_g, which equals $\phi''(T_g) > 0$ by the strict convexity of ϕ. $\square$

The content of Proposition 1 is that conditioning on regime state does not destroy the propriety of scoring rules, provided the conditional distributions remain within the class $\mathcal{F}$ over which consistency was established. This condition is mild: if $\mathcal{F}$ is the class of distributions with finite second moments, then any regime-conditional distribution inheriting finite second moments satisfies the requirement.

The proposition has an important consequence for model evaluation. If we wish to compare two volatility forecasting models within a specific regime, we can use the same scoring function (QLIKE, squared error, or any other strictly consistent score) applied to the subsample of observations falling in that regime, and the resulting ranking will be a valid comparison of the models' regime-conditional performance. There is no need to construct special "regime-aware" scoring functions; standard scoring functions retain their desirable properties when conditioned on regime state.

However, the fact that conditional rankings are valid within each regime does not imply that they agree with unconditional rankings. The unconditional expected loss is a weighted

average of conditional expected losses,

$$\mathbb{E}[S(\hat{y}, Y)] = \sum_{g \in \mathcal{G}} \pi_g \, \mathbb{E}[S(\hat{y}, Y) \mid G = g],$$

where $\pi_g = \mathbb{P}(G = g)$ is the unconditional probability of regime g. When models differ in their regime-conditional performance and regimes have unequal probabilities, the unconditional ranking is dominated by the model's performance in the most frequent regime. This observation motivates the ranking reversal analysis of the next section.

2.3 Ranking Reversals Under Regime Imbalance

We now formalize the conditions under which unconditional loss rankings can reverse relative to regime-conditional rankings. This result provides the central theoretical motivation for regime-conditional evaluation.

We note at the outset that the algebraic content of the results in this section is straightforward; their value lies in formalizing the conditions under which unconditional evaluation fails for regime-switching models and in quantifying the relationship between regime imbalance and ranking reversal.

Consider two competing forecasting models, A and B, and a strictly consistent scoring function S. Define the regime-conditional expected loss of model $m \in \{A, B\}$ under regime g as

$$L_m(g) = \mathbb{E}[S(\hat{y}_m, Y) \mid G = g],$$

and the regime-conditional loss differential as

$$d(g) = L_A(g) - L_B(g).$$

The unconditional loss differential is

$$\bar{d} = \mathbb{E}[S(\hat{y}_A, Y)] - \mathbb{E}[S(\hat{y}_B, Y)] = \sum_{g \in \mathcal{G}} \pi_g \, d(g).$$

The unconditional ranking agrees with the conditional ranking in every regime if and only if $d(g)$ has the same sign for all $g \in \mathcal{G}$. When $d(g)$ changes sign across regimes—model A is better in some regimes and worse in others—the unconditional ranking depends on the regime probabilities π_g, and ranking reversals become possible.

Definition 3 (Ranking reversal). A ranking reversal occurs when model A has lower unconditional loss than model B ($\bar{d} < 0$) but model B has lower conditional loss in at least one regime g^* ($d(g^*) > 0$). A ranking reversal *with respect to a pre-specified regime* g^* occurs when $\bar{d} < 0$ but $d(g^*) > 0$ for a regime g^* designated a priori as the regime of greatest economic interest (e.g., the high-volatility stress regime for risk management applications).

We now establish the conditions under which ranking reversals occur.

Theorem 1 (Ranking reversal under regime imbalance). Consider a two-regime setting with $\mathcal{G} = \{L, H\}$ (low and high volatility), regime probabilities π_L and $\pi_H = 1 - \pi_L$, and a strictly consistent scoring function S. Let models A and B produce forecasts with regime-conditional loss differentials $d(L) = L_A(L) - L_B(L)$ and $d(H) = L_A(H) - L_B(H)$.

Suppose model A is a regime-switching model that is better in the high-volatility regime but worse in the low-volatility regime, while model B is a simpler model that is uniformly adequate:

$$d(L) > 0 \quad \text{and} \quad d(H) < 0.$$

Then the following hold:

(i) Model B is ranked as the unconditional winner (i.e., $\bar{d} > 0$, so A loses unconditionally) if and only if

$$\pi_L \, d(L) + \pi_H \, d(H) > 0 \qquad \Longleftrightarrow \qquad \frac{\pi_L}{\pi_H} > \frac{-d(H)}{d(L)} = \frac{|d(H)|}{d(L)}.$$

(ii) A ranking reversal occurs—B wins unconditionally while A wins in the high-volatility regime—whenever the regime imbalance ratio satisfies

$$\frac{\pi_L}{\pi_H} > \frac{|d(H)|}{d(L)}.$$

(iii) For the empirically relevant case where $d(L) \approx |d(H)|$ (the regime-switching model's gain in stress is comparable in magnitude to its cost in calm), a ranking reversal occurs whenever $\pi_L > \pi_H$, that is, whenever the low-volatility regime is more frequent than the high-volatility regime.

(iv) *Finite-sample probability bound.* Let $\hat{d}(g)$ denote the sample loss differential in regime g, with $\mathrm{Var}(\hat{d}(g)) = \sigma_g^2/n_g$ where n_g is the number of observations in regime g. If the population satisfies $d(H) < 0 < d(L)$ and $\bar{d} > 0$ (i.e., B wins unconditionally),

then for any sample of size n the probability that the sample unconditional ranking correctly reflects the unconditional population ranking satisfies

$$\mathbb{P}(\hat{\bar{d}} > 0) \geq 1 - \exp\left(-\frac{n\bar{d}^2}{2(\pi_L \sigma_L^2 + \pi_H \sigma_H^2)}\right),$$

under sub-Gaussian tail assumptions on the loss differential process. Meanwhile, the probability that the sample detects the reversal in the stress regime (i.e., that $\hat{d}(H) < 0$) satisfies the analogous bound with $n_H = n\pi_H$ replacing n. Since $n_H \ll n$ when $\pi_H \ll 1$, substantially larger samples are needed to detect conditional superiority than unconditional superiority.

Proof. The unconditional loss differential decomposes as

$$\bar{d} = \pi_L \, d(L) + \pi_H \, d(H).$$

Since $d(L) > 0$ (model A is worse in calm) and $d(H) < 0$ (model A is better in stress), the sign of $\bar{d}$ depends on the relative magnitudes of the two terms. Setting $\bar{d} > 0$:

$$\pi_L \, d(L) + \pi_H \, d(H) > 0$$

$$\pi_L \, d(L) > -\pi_H \, d(H) = \pi_H \, |d(H)|$$

$$\frac{\pi_L}{\pi_H} > \frac{|d(H)|}{d(L)}.$$

This establishes part (i). Part (ii) is a direct restatement: the condition $\bar{d} > 0$ (B wins

unconditionally) combined with $d(H) < 0$ (A wins in high regime) constitutes a ranking reversal by Definition 3.

For part (iii), when $d(L) \approx |d(H)|$, the right-hand side of the inequality is approximately one. Therefore the condition reduces to $\pi_L/\pi_H > 1$, or equivalently $\pi_L > 1/2$. Since in financial markets the low-volatility regime typically occupies 70–85% of observations (Hamilton 1989; Guidolin and Timmermann 2007), the condition $\pi_L > \pi_H$ is almost always satisfied.

For part (iv), the sample unconditional loss differential $\hat{\bar{d}} = n^{-1} \sum_{t=1}^{n} \Delta L_t$ is a weighted sum of within-regime sample means. Under sub-Gaussian assumptions, Hoeffding's inequality applied to the loss differential process gives $\mathbb{P}(\hat{\bar{d}} \leq 0) \leq \exp(-n\bar{d}^2/(2V))$ where $V = \pi_L \sigma_L^2 + \pi_H \sigma_H^2$ is the unconditional variance of the loss differential. The analogous bound for $\hat{d}(H)$ replaces n with the effective sample size $n_H = n\pi_H$. The ratio $n/n_H = 1/\pi_H$ quantifies the additional sample size needed for conditional inference. □

Theorem 1 reveals a systematic bias in unconditional forecast evaluation: when market regimes are imbalanced, as they invariably are in financial data, unconditional loss functions overweight performance during calm periods. A regime-switching model that sacrifices a modest amount of accuracy during calm periods to achieve substantially better forecasts during stress may be rejected by unconditional evaluation simply because calm periods dominate the sample. Part (iv) quantifies this effect: detecting conditional superiority in the stress regime requires a sample roughly $1/\pi_H$ times larger than detecting unconditional differences, which for $\pi_H \approx 0.2$ implies a five-fold increase in sample size requirements.

The result extends naturally to the K-regime case. Let $\mathcal{G} = \{1, \dots, K\}$ with probabilities $\pi_1, \dots, \pi_K$.

Corollary 1 (Multi-regime ranking reversal). In the K-regime setting, a ranking reversal with respect to regime g^* occurs whenever

$$\sum_{g \neq g^*} \pi_g \, d(g) > \pi_{g^*} \, |d(g^*)|.$$

That is, the cumulative disadvantage of model A in all other regimes, weighted by their probabilities, outweighs model A's advantage in the regime of interest. This condition is increasingly easy to satisfy as π_{g^*} decreases, implying that ranking reversals are most likely for rare but economically important regimes.

The practical implication of Theorem 1 and its corollary is that unconditional QLIKE rankings carry a built-in bias against regime-switching models in precisely the setting where such models are most needed: when the regime of interest (crisis, high volatility, systemic stress) is rare relative to the dominant regime (calm, normal markets). This provides a theoretical rationale for regime-conditional evaluation.

2.4 Conditions for Agreement Between Unconditional and Conditional Rankings

While Section 2.3 established conditions for ranking reversal, it is equally important to characterize conditions under which unconditional evaluation is reliable. This section provides sufficient conditions for unconditional and conditional rankings to agree across all regimes, thereby identifying settings where standard evaluation remains valid.

Theorem 2 (Conditions for ranking agreement). Let S be a strictly consistent scoring

function, $\mathcal{G} = \{1, \dots, K\}$ a finite set of regimes with positive probabilities $\pi_g > 0$ for all g, and $d(g) = L_A(g) - L_B(g)$ the regime-conditional loss differential between models A and B. Then the unconditional ranking agrees with the conditional ranking in every regime under any of the following sufficient conditions:

(C1) *Uniform dominance:* $d(g) \leq 0$ for all $g \in \mathcal{G}$ (or $d(g) \geq 0$ for all g). That is, one model is weakly superior in every regime.

(C2) *Balanced regimes with bounded differentials:* When regime frequencies are balanced ($\pi_g = 1/K$ for all g), the unconditional differential equals the unweighted average of conditional differentials:

$$\bar{d} = \frac{1}{K} \sum_{g=1}^{K} d(g).$$

If additionally the magnitude $|d(g)|$ is bounded by some constant M for all g, then the unconditional ranking agrees with the modal conditional ranking whenever a strict majority of regimes favor the same model. Specifically, if $|\{g : d(g) < 0\}| > K/2$ and $|d(g)| \leq M$ for all g, then $\bar{d} < 0$.

(C3) *Proportional loss differentials:* There exists a constant $\lambda > 0$ such that $|d(g)| = \lambda$ for all g, and the set $\{g : d(g) < 0\}$ has total probability exceeding $1/2$. That is, when loss differentials are equal in magnitude across regimes, the unconditional winner is determined by whether the model wins in more than half the probability mass.

Proof. We prove each condition separately.

Condition (C1). If $d(g) \leq 0$ for all g, then

$$\bar{d} = \sum_{g \in \mathcal{G}} \pi_g \, d(g) \leq 0$$

since each term is non-positive and $\pi_g > 0$. Moreover, if $d(g) < 0$ for at least one g with $\pi_g > 0$, then $\bar{d} < 0$ strictly. In this case, model A wins both unconditionally and in every regime, so no ranking reversal can occur. The symmetric argument applies when $d(g) \geq 0$ for all g.

Condition (C2). When $\pi_g = 1/K$ for all g,

$$\bar{d} = \frac{1}{K} \sum_{g=1}^{K} d(g).$$

Suppose a strict majority of regimes favor model A: let $|\{g : d(g) < 0\}| = J > K/2$. When $|d(g)| \leq M$ for all g, each term in the sum is bounded below by $-M$ and above by M. The sum over the J favorable regimes contributes at most $-J \cdot \epsilon$ for some $\epsilon > 0$ (since each such $d(g) < 0$), while the sum over the remaining $K - J$ unfavorable regimes contributes at most $(K - J) \cdot M$. However, unlike the equal-magnitude case of (C3), the average of the differentials need not have the same sign as the majority: when magnitudes differ, a minority of regimes with large differentials can outweigh a majority with small differentials. The boundedness condition $|d(g)| \leq M$ limits but does not eliminate this possibility. Full agreement between the unconditional ranking and the modal conditional ranking therefore requires either equal magnitudes across regimes (reducing to (C3)) or the stronger condition that $\sum_{g=1}^{K} d(g) \neq 0$ with the appropriate sign. The condition $\sum_{g=1}^{K} d(g) \neq 0$ ensures a

determinate ranking.

Condition (C3). When $|d(g)| = \lambda$ for all g, define $\mathcal{G}^+ = \{g : d(g) > 0\}$ and $\mathcal{G}^- = \{g : d(g) < 0\}$. Then

$$\bar{d} = \lambda \sum_{g \in \mathcal{G}^+} \pi_g - \lambda \sum_{g \in \mathcal{G}^-} \pi_g = \lambda \left(\pi(\mathcal{G}^+) - \pi(\mathcal{G}^-) \right),$$

where $\pi(\mathcal{G}^\pm) = \sum_{g \in \mathcal{G}^\pm} \pi_g$. If $\pi(\mathcal{G}^-) > 1/2$, then $\bar{d} < 0$ and the unconditional ranking agrees with the conditional ranking for the majority of the probability mass. In this case, A wins unconditionally and wins in regimes carrying more than half the probability. □

Condition (C1) is the most practically informative: unconditional evaluation is reliable when one model dominates across all regimes. This corresponds to the case where a model is genuinely and uniformly better, not merely better on average. In the volatility forecasting context, (C1) holds when a model improves upon the benchmark in both calm and stress periods. A model that achieves this uniform dominance should be preferred by any reasonable evaluation criterion, unconditional or conditional.

Condition (C2) highlights that the problem identified in Theorem 1 is fundamentally one of regime imbalance, not of improper scoring. When regimes are balanced, the unconditional average is a fair summary of average behavior, though it does not preclude reversals in individual regimes when magnitudes differ. This condition is almost never satisfied in financial data, where calm periods vastly outnumber stress episodes, but it serves as a theoretical benchmark.

Condition (C3) addresses the special case of symmetric gains and losses across regimes. When the absolute improvement in one regime equals the absolute deterioration in another, the

unconditional ranking is determined by which regimes carry more probability mass. This condition formalizes the intuition that regime-switching models face an uphill battle in unconditional evaluation: to win overall, they must improve in regimes that collectively exceed 50% of the sample.

Remark 1 (Diagnostic for ranking reliability). The quantity

$$R = \frac{\max_g d(g) - \min_g d(g)}{|\bar{d}|}$$

provides a descriptive diagnostic for the severity of regime heterogeneity in loss differentials relative to the overall signal. When R is large (loss differentials vary substantially across regimes relative to their average), unconditional rankings are fragile and may not reflect conditional performance. When R is small (loss differentials are similar across regimes), unconditional rankings are robust. This ratio serves as a practical heuristic for whether regime-conditional evaluation is warranted. We note that R is a sample-dependent statistic whose sampling distribution depends on the joint distribution of loss differentials across regimes. We investigate the sampling properties of R through simulation in Section 3, where we calibrate critical values under various data-generating processes.

2.5 Regime-Conditional Model Confidence Set

The Model Confidence Set (MCS) procedure of Hansen et al. (2011) provides a systematic approach to model selection that accounts for estimation uncertainty. The MCS identifies a set $\hat{\mathcal{M}}^*_\alpha$ of models that are not significantly outperformed at level α by any other model in

the comparison set. This section extends the MCS to regime-conditional settings.

Let $\mathcal{M} = \{1, \dots, M\}$ denote a set of competing forecasting models. For models $i, j \in \mathcal{M}$, define the regime-conditional loss differential process

$$d_{ij,t}(g) = S(\hat{y}_{i,t}, y_t) - S(\hat{y}_{j,t}, y_t), \quad \text{for } t \text{ such that } G_t = g,$$

and the population regime-conditional mean differential

$$\mu_{ij}(g) = \mathbb{E}[d_{ij,t}(g)] = L_i(g) - L_j(g).$$

The regime-conditional MCS is the set of models that are not significantly outperformed in regime g.

Definition 4 (Regime-conditional Model Confidence Set). For regime $g \in \mathcal{G}$ and significance level $\alpha \in (0, 1)$, the regime-conditional superior set of models is

$$\mathcal{M}_g^* = \{i \in \mathcal{M} : \mu_{ij}(g) \leq 0 \text{ for all } j \in \mathcal{M}\},$$

consisting of models whose conditional expected loss in regime g is not exceeded by any other model. The regime-conditional Model Confidence Set $\hat{\mathcal{M}}_{g,\alpha}^*$ is a data-dependent set constructed to satisfy

$$\lim_{n \to \infty} \mathbb{P}\left(\mathcal{M}_g^* \subseteq \hat{\mathcal{M}}_{g,\alpha}^*\right) \geq 1 - \alpha.$$

The construction of $\hat{\mathcal{M}}_{g,\alpha}^*$ follows the sequential elimination procedure of Hansen

et al. (2011), applied to the subsample of observations falling in regime g. Define $n_g = \sum_{t=1}^{n} \mathbf{1}(G_t = g)$ as the number of observations in regime g, and let $\bar{d}_{ij}(g) = n_g^{-1} \sum_{t:G_t=g} d_{ij,t}(g)$ be the sample mean loss differential within regime g.

The equivalence hypothesis for regime g is

$$H_{0,\mathcal{M},g} : \mu_{ij}(g) = 0 \quad \text{for all } i, j \in \mathcal{M}.$$

We test this using the range statistic adapted to the regime-conditional setting:

$$T_{R,g} = \max_{i,j \in \mathcal{M}} \frac{|\bar{d}_{ij}(g)|}{\sqrt{\widehat{\mathrm{Var}}(\bar{d}_{ij}(g))}},$$

where $\widehat{\mathrm{Var}}(\bar{d}_{ij}(g))$ is a HAC-consistent variance estimator applied to the regime-g subsample. The sequential elimination proceeds as in the standard MCS: if the null of equal conditional predictive ability in regime g is rejected, the worst-performing model (the model i that maximizes $\bar{d}_{i\cdot}(g) = (M-1)^{-1} \sum_{j \neq i} \bar{d}_{ij}(g)$) is removed, and the test is repeated on the reduced set.

Proposition 2 (Validity of regime-conditional MCS). Assume the following conditions hold:

(A1) The joint process $\{(Y_t, G_t, \hat{y}_{1,t}, \dots, \hat{y}_{M,t})\}$ is strictly stationary and α-mixing with mixing coefficients satisfying $\sum_{k=1}^{\infty} \alpha(k)^{\delta/(2+\delta)} < \infty$ for some $\delta > 0$.

(A2) The effective sample size $n_g \to \infty$ as $n \to \infty$, which requires $\pi_g > 0$.

(A3) A functional central limit theorem applies to $n_g^{-1/2} \sum_{t:G_t=g} (d_{ij,t}(g) - \mu_{ij}(g))$.

Then the regime-conditional MCS $\widehat{\mathcal{M}}^*_{g,\alpha}$ has correct asymptotic coverage:

$$\lim_{n\to\infty} \mathbb{P}\left(\mathcal{M}^*_g \subseteq \widehat{\mathcal{M}}^*_{g,\alpha}\right) \geq 1-\alpha.$$

Proof. The proof proceeds by verifying that the conditions of Theorem 1 in Hansen et al. (2011) are satisfied for the regime-conditional subsample. We first address the mixing properties of the regime-conditional subprocess.

Under (A1), the regime-conditional loss differential process $\{d_{ij,t}(g)\}_{t\in\mathcal{T}_g}$, where $\mathcal{T}_g = \{t : G_t = g\}$, inherits mixing properties from the full process. Specifically, the subsequence $\{X_{t_j}\}$ indexed by the regime-g times $t_1 < t_2 < \cdots$ preserves α-mixing because the regime indicator process $\{G_t\}$ is itself stationary and mixing under (A1). This follows from a general result on mixing subsequences: if $\{X_t\}$ is α-mixing and $\{t_j\}$ is a subsequence determined by a stationary, mixing selection process, then the subsequence $\{X_{t_j}\}$ preserves α-mixing with mixing coefficients dominated by those of the original process (Bradley 2007, Theorem 3.49). The key requirement is that the selection mechanism (here, $G_t = g$) does not introduce long-range dependence, which is guaranteed by the mixing assumption on the joint process in (A1).

Under (A2), the regime-conditional subsample grows proportionally with n at rate $\pi_g n$, ensuring a divergent effective sample size.

Under (A3), the regime-conditional sample mean $\bar{d}_{ij}(g)$ satisfies a central limit theorem. This CLT for the subsampled process holds under the inherited mixing conditions by applying the CLT for α-mixing sequences (Ibragimov 1962; Merlevede et al. 2011). The asymptotic

variance is consistently estimated by the HAC estimator applied to the regime-g subsample.

Given these conditions, the equivalence test based on $T_{R,g}$ has correct asymptotic size, and the sequential elimination procedure controls the familywise error rate at level α by the same step-down argument as in Hansen et al. (2011). The critical values are obtained by block bootstrap resampling within the regime-g subsample, which consistently estimates the null distribution under (A1)–(A3). □

The regime-conditional MCS framework provides several diagnostic insights not available from unconditional MCS analysis. First, a model may belong to $\widehat{\mathcal{M}}^*_{L,\alpha}$ (the set of models not rejected in the calm regime) but not to $\widehat{\mathcal{M}}^*_{H,\alpha}$ (rejected in the stress regime), revealing it as a "calm-period specialist." Second, a model in $\widehat{\mathcal{M}}^*_{H,\alpha}$ but not $\widehat{\mathcal{M}}^*_{L,\alpha}$ is a "stress specialist." Third, a model in the intersection $\widehat{\mathcal{M}}^*_{L,\alpha} \cap \widehat{\mathcal{M}}^*_{H,\alpha}$ is an "all-weather" model. This taxonomy is informative for practitioners whose loss functions may weight stress-period performance more heavily than calm-period performance.

A practical consideration is the effective sample size within each regime. Since stress periods are rare, n_H may be small even when n is large, reducing the power of the MCS procedure in the high-volatility regime. This is an unavoidable feature of the data—the same rarity that causes unconditional metrics to underweight stress performance also limits the precision of stress-conditional inference. We address this limitation empirically by reporting both asymptotic and bootstrap p-values and by conducting power analysis.

2.6 Extension to Joint VaR/ES Elicitability Under Regime Conditioning

The Fissler-Ziegel framework (Fissler and Ziegel 2016) established that while Value-at-Risk (VaR) is elicitable on its own and Expected Shortfall (ES) is not, the pair $(\mathrm{VaR}_\alpha, \mathrm{ES}_\alpha)$ is jointly elicitable. Specifically, there exists a strictly consistent scoring function for the bivariate functional $T(F) = (\mathrm{VaR}_\alpha(F), \mathrm{ES}_\alpha(F))$. The canonical representative of this class, known as the FZ0 loss, is

$$S_{\mathrm{FZ0}}(\hat{v}, \hat{e}, y) = \frac{1}{\hat{e}}\left(\hat{v} - y + \frac{(y-\hat{v})^+}{\alpha}\right) - \frac{1}{\hat{e}} + \log(-\hat{e}),$$

where $\hat{v}$ and $\hat{e}$ are the VaR and ES forecasts, y is the realized return, α is the probability level, $(x)^+ = \max(x, 0)$, and $\hat{e} < 0$ denotes the ES at level α for the left tail (i.e., $\hat{e} = \mathbb{E}[Y \mid Y \leq \hat{v}]$ under the forecast distribution, which is negative for loss quantiles).

We extend the regime-conditioning framework to the FZ0 loss. The regime-conditional risk measures are defined as

$$\mathrm{VaR}_{\alpha,g} = T_1(F_{Y|G=g}), \quad \mathrm{ES}_{\alpha,g} = T_2(F_{Y|G=g}),$$

where T_1 and T_2 extract the VaR and ES from the regime-conditional return distribution.

Proposition 3 (Regime-conditional joint elicitability of VaR and ES). Let S_{FZ0} be the Fissler-Ziegel scoring function that is strictly consistent for $(\mathrm{VaR}_\alpha, \mathrm{ES}_\alpha)$ relative to a class $\mathcal{F}$ of distributions with finite first moments. If $F_{Y|G=g} \in \mathcal{F}$ for each $g \in \mathcal{G}$, then S_{FZ0} is

strictly consistent for $(\mathrm{VaR}_{\alpha,g}, \mathrm{ES}_{\alpha,g})$ under $F_{Y|G=g}$.

That is, the regime-conditional expected FZ0 loss,

$$L_{\mathrm{FZ0}}(g; \hat{v}, \hat{e}) = \mathbb{E}\left[S_{\mathrm{FZ0}}(\hat{v}, \hat{e}, Y) \mid G = g\right],$$

is uniquely minimized at $(\hat{v}, \hat{e}) = (\mathrm{VaR}_{\alpha,g}, \mathrm{ES}_{\alpha,g})$.

Proof. The argument is identical in structure to Proposition 1. The joint elicitability result of Fissler and Ziegel (2016) establishes strict consistency of S_{FZ0} for all distributions in $\mathcal{F}$. Since $F_{Y|G=g} \in \mathcal{F}$ by assumption, the conditional expected score is minimized at the conditional risk measures. The multi-dimensional nature of the functional does not affect the conditioning argument: the minimization over the pair $(\hat{v}, \hat{e})$ under the conditional distribution proceeds exactly as under the unconditional distribution. □

The practical import of Proposition 3 for risk management is substantial. A regime-switching model that calibrates VaR and ES accurately during stress but poorly during calm periods would, under unconditional FZ0 evaluation, be penalized for its calm-period miscalibration. Since calm periods dominate the sample, this model may rank below a simpler model that provides mediocre but uniform calibration. Yet from a risk management perspective, stress-period calibration is far more consequential: VaR violations during calm markets are painful but manageable, while VaR violations during stress can trigger systemic consequences through margin calls, forced liquidation, and regulatory breaches.

The ranking reversal result of Theorem 1 applies directly to the FZ0 loss. Replacing S with S_{FZ0} and defining $d_{\mathrm{FZ0}}(g)$ as the regime-conditional FZ0 loss differential, the conditions

for ranking reversal are identical. A risk management application of regime-conditional evaluation would compare models using $L_{\mathrm{FZ0}}(H; \hat{v}, \hat{e})$—the expected FZ0 loss during stress—as the primary criterion, with unconditional FZ0 serving as a secondary diagnostic.

Corollary 2 (Regime-conditional risk evaluation). The ranking reversal conditions of Theorem 1, applied to S_{FZ0}, imply that unconditional backtesting of VaR/ES models systematically favors models that calibrate well during calm periods. For regulatory applications where stress-period performance is paramount, regime-conditional FZ0 evaluation provides a more relevant ranking.

2.7 Connection to Giacomini-White Conditional Predictive Ability

The conditional predictive ability (CPA) framework of Giacomini and White (2006) provides a general approach to testing whether two forecasting models have equal predictive accuracy conditional on an information set $\mathcal{H}_t$ available at time t. Their test statistic is based on the moment condition

$$\mathbb{E}[\Delta L_t \cdot h_t] = 0,$$

where $\Delta L_t = S(\hat{y}_{A,t}, y_t) - S(\hat{y}_{B,t}, y_t)$ is the loss differential and h_t is a vector of test functions measurable with respect to $\mathcal{H}_t$. When $h_t = \mathbf{1}(G_t = g)$ for some regime indicator G_t, the test reduces to comparing conditional expected losses in regime g, which is precisely the regime-conditional comparison developed in Sections 2.2–2.4.

Our framework differs from Giacomini and White (2006) in two important respects. First, the Giacomini-White test conditions on observable instruments h_t that are in the forecaster's

information set. When the regime variable G_t is observable (e.g., an indicator based on the VIX exceeding a threshold), the regime-conditional test is a direct application of their framework with $h_t = \mathbf{1}(G_t = g)$. In this case, the Giacomini-White test and our regime-conditional comparison are equivalent, and the asymptotic theory follows directly from their results.

Second, and more substantively, our setting involves conditioning on a *latent* regime state that is estimated from the data. Let G_t^* denote the true (latent) regime state and $\hat{G}_t$ denote its estimate (e.g., from a VRNN or a Markov-switching filter). The regime-conditional loss using the estimated regime is

$$\hat{L}_m(g) = \mathbb{E}[S(\hat{y}_m, Y) \mid \hat{G}_t = g],$$

which differs from the target $L_m(g) = \mathbb{E}[S(\hat{y}_m, Y) \mid G_t^* = g]$ due to regime estimation error. This introduces an additional source of uncertainty not present in the Giacomini-White framework, which treats the conditioning variables as known.

Proposition 4 (Asymptotic validity under consistent regime estimation). Let $\hat{G}_t$ be an estimate of the latent regime state G_t^* satisfying the following conditions:

(B1) *Consistency:* $\mathbb{P}(\hat{G}_t \neq G_t^*) \to 0$ as the estimation sample grows.

(B2) *Uniform convergence:* $\sup_{g \in \mathcal{G}} |\hat{\pi}_g - \pi_g| \xrightarrow{p} 0$, where $\hat{\pi}_g = n^{-1} \sum_{t=1}^{n} \mathbf{1}(\hat{G}_t = g)$.

(B3) *Asymptotic negligibility of misclassification:* $n^{1/2}(M_n/n) \to 0$ in probability, where $M_n = \sum_{t=1}^{n} \mathbf{1}(\hat{G}_t \neq G_t^*)$ is the total misclassification count.

Then the regime-conditional loss estimate based on $\hat{G}_t$,

$$\hat{L}_m(g) = \frac{1}{\hat{n}_g} \sum_{t:\hat{G}_t=g} S(\hat{y}_{m,t}, y_t),$$

is consistent for $L_m(g)$, and the regime-conditional Giacomini-White test statistic using $h_t = \mathbf{1}(\hat{G}_t = g)$ has the same asymptotic distribution as the infeasible test using $h_t = \mathbf{1}(G_t^* = g)$.

Proof. We decompose the error into a classification component and an estimation component. Write $\hat{n}_g = \sum_{t=1}^n \mathbf{1}(\hat{G}_t = g)$ and $n_g^* = \sum_{t=1}^n \mathbf{1}(G_t^* = g)$. Consider the decomposition

$$\hat{L}_m(g) - L_m(g) = \underbrace{\left(\hat{L}_m(g) - \tilde{L}_m(g)\right)}_{\text{classification error}} + \underbrace{\left(\tilde{L}_m(g) - L_m(g)\right)}_{\text{estimation error}},$$

where $\tilde{L}_m(g) = (n_g^*)^{-1} \sum_{t:G_t^*=g} S(\hat{y}_{m,t}, y_t)$ is the infeasible regime-conditional loss using true regime labels.

The estimation error term converges to zero in probability by the law of large numbers applied to the stationary process $\{S(\hat{y}_{m,t}, y_t) \cdot \mathbf{1}(G_t^* = g)\}$, under the stationarity and mixing conditions assumed throughout.

For the classification error, the total misclassification count is $M_n = \sum_{t=1}^n \mathbf{1}(\hat{G}_t \neq G_t^*)$, and each misclassification perturbs the sum by at most $\|S\|_\infty$ (assuming bounded scores; see Remark 2 below for the extension to unbounded scores). Under (B1), $M_n/n \xrightarrow{p} 0$, so

$$\left|\hat{L}_m(g) - \tilde{L}_m(g)\right| \leq \frac{2M_n \|S\|_\infty}{\hat{n}_g},$$

which converges to zero in probability since $\hat{n}_g/n \xrightarrow{p} \pi_g > 0$ by (B2).

For the distributional result, the regime-conditional test statistic is

$$\hat{t}_g = \frac{\bar{d}(g)}{\sqrt{\widehat{\mathrm{Var}}(\bar{d}(g))/\hat{n}_g}},$$

where $\bar{d}(g) = \hat{L}_A(g) - \hat{L}_B(g)$. Under condition (B3), the contribution of misclassified observations to $\sqrt{\hat{n}_g}\,\bar{d}(g)$ is $o_p(1)$, since $M_n = o_p(n^{1/2})$ by (B3) and the scores are bounded. Therefore, $\sqrt{\hat{n}_g}\,\bar{d}(g)$ has the same limiting distribution as the infeasible statistic $\sqrt{n_g^*}\,\tilde{d}(g)$, which is asymptotically normal under the mixing and CLT conditions of (A1)–(A3) from Proposition 2. □

It is important to distinguish two sources of misclassification in condition (B3). *Estimation-induced misclassification* arises from the finite-sample error in estimating the regime classifier and vanishes asymptotically under (B1). *Population misclassification* arises when the true data-generating process does not admit a perfect classification rule—for instance, when the regime states have overlapping conditional distributions so that the Bayes-optimal classifier has positive error rate. When population misclassification is positive, M_n/n converges to a positive constant, and condition (B3) fails. In this case, the regime-conditional test remains valid as a test of conditional predictive ability given the *estimated* regime $\hat{G}_t$, analogous to the Giacomini-White test with a noisy but informative instrument. The test has correct size for the null hypothesis $\mathbb{E}[\Delta L_t \mid \hat{G}_t = g] = 0$ and retains power against alternatives where models differ conditional on the estimated regime, even though the estimated regime may not perfectly track the latent state.

Remark 2 (Extension to unbounded scores). The proof of Proposition 4 uses bounded

scores to control the classification error term. The QLIKE loss $S_{\text{QLIKE}}(\hat{\sigma}^2, \sigma^2) = \sigma^2/\hat{\sigma}^2 - \log(\sigma^2/\hat{\sigma}^2) - 1$ is unbounded, since it diverges as $\hat{\sigma}^2 \to 0$ or as $\sigma^2/\hat{\sigma}^2 \to \infty$. The extension to unbounded scores is straightforward under finite moment conditions. Specifically, if $\mathbb{E}[|S(\hat{y}_{m,t}, y_t)|^{2+\delta}] < \infty$ for some $\delta > 0$, then the classification error bound becomes

$$\left|\hat{L}_m(g) - \tilde{L}_m(g)\right| \leq \frac{2M_n}{\hat{n}_g} \left(\frac{1}{M_n} \sum_{t \in \mathcal{M}_n} |S(\hat{y}_{m,t}, y_t)| \right),$$

where $\mathcal{M}_n = \{t : \hat{G}_t \neq G_t^*\}$ is the set of misclassified observations. Under the moment condition, the average score over misclassified observations is $O_p(1)$, and the bound reduces to $O_p(M_n/\hat{n}_g) = o_p(1)$. For the distributional result, the contribution of misclassified observations to $\sqrt{\hat{n}_g}\,\bar{d}(g)$ is controlled by $M_n/n^{1/2}$ times the average score magnitude, which is $o_p(1)$ under (B3) and the finite fourth moment condition $\mathbb{E}[\Delta L_t^4] < \infty$.

For parametric regime models such as Markov-switching, the misclassification probability decreases exponentially in the regime persistence parameter and satisfies (B3) under standard regularity conditions (cf. Douc et al. 2004). For the VRNN-based regime classifier used in our empirical application, verifying (B3) analytically requires parametric assumptions on the data-generating process that we are not prepared to impose. We therefore treat (B3) as a maintained assumption and rely on block bootstrap inference over both the regime estimation and forecast comparison stages as the operational approach for valid inference; see the discussion below.

Generated-regressor uncertainty. When condition (B3) fails—or more generally, when the regime estimation error makes a non-negligible contribution to the sampling distribution of the test statistic—the regime-conditional comparison inherits a generated-regressor prob-

lem. The estimated regime $\hat{G}_t$ enters the test as if it were observed, but it is itself a function of the data, introducing additional variability that is not captured by standard asymptotic standard errors. In this setting, standard errors computed from the second-stage forecast comparison alone may be too small, leading to over-rejection of the null hypothesis of equal conditional predictive ability.

The appropriate inferential strategy is a block bootstrap that resamples over both stages of the procedure: regime estimation and forecast comparison. Specifically, one draws bootstrap samples of contiguous blocks from the original time series, re-estimates the regime classifier on each bootstrap sample, re-computes the regime-conditional loss differentials, and constructs bootstrap confidence intervals for the test statistic. This two-stage bootstrap provides valid inference by implicitly accounting for the joint sampling distribution of the regime estimates and the loss differentials (Goncalves and White 2004). In our empirical implementation, we employ the circular block bootstrap with block length chosen by the rule of Politis and White (2004), applied to the full two-stage procedure. This approach is conservative in the sense that it captures both regime estimation uncertainty and forecast evaluation uncertainty, at the cost of somewhat wider confidence intervals relative to the infeasible procedure with known regimes.

The connection to Giacomini and White (2006) also illuminates the relationship between regime-conditional evaluation and unconditional evaluation. In their framework, the unconditional Diebold-Mariano test is a special case where $h_t = 1$ (no conditioning). The regime-conditional test with $h_t = \mathbf{1}(\hat{G}_t = g)$ is a strict refinement: it can detect differences in predictive ability that are masked by the unconditional test when model superiority varies

across regimes. The unconditional test can reject the null of equal predictive ability only when the average loss differential is non-zero; the regime-conditional test can reject when the loss differential is non-zero in a specific regime, even if it averages to zero unconditionally.

Corollary 3 (Refinement of Diebold-Mariano). Let $\bar{d} = \mathbb{E}[\Delta L_t]$ denote the unconditional mean loss differential and $d(g) = \mathbb{E}[\Delta L_t \mid G_t^* = g]$ the regime-conditional mean loss differential. If there exist regimes g_1, g_2 with $d(g_1) < 0 < d(g_2)$ and $\sum_g \pi_g d(g) = 0$, then the unconditional Diebold-Mariano test has no power to detect the difference in predictive ability (since $\bar{d} = 0$), but the regime-conditional test rejects the null of equal conditional predictive ability in each regime for which $d(g) \neq 0$, provided the subsample is sufficiently large. Regime-conditional evaluation is therefore a strict refinement of unconditional evaluation in the sense that it can detect differences invisible to the unconditional test.

3. Simulation Study

The theoretical results of Section 2 establish conditions under which unconditional scoring rules produce ranking reversals for regime-switching volatility models. This section investigates the practical frequency and severity of such reversals through a controlled Monte Carlo experiment. The simulation serves three purposes: to verify that Theorems 1 and 2 correctly predict the occurrence and absence of ranking reversals under known data-generating processes; to quantify how the misranking frequency varies with regime imbalance, volatility contrast, and sample size; and to establish the size and power properties of the regime-conditional Diebold-Mariano test relative to its unconditional counterpart.

3.1 Data-Generating Processes

We specify three data-generating processes (DGPs) that span the range of structures relevant for volatility forecasting. The first is a discrete Markov-switching process with known regime states, representing the idealized case in which regime-conditional evaluation can be performed without estimation error. The second is a smooth transition process with a continuous regime index, representing the more realistic setting of the STR-HAR framework developed in Lim (2026a). The third is a single-regime process that serves as a null benchmark in which no regime structure exists and unconditional evaluation should correctly rank all models.

DGP 1: Markov-Switching (MS). The log realized variance follows a two-state Markov-switching process. In the calm regime ($G_t = L$), the log variance is drawn as $\log \sigma_t^2 = \mu_L + \omega_L \eta_t$ with $\eta_t \sim N(0,1)$. In the stress regime ($G_t = H$), the log variance is $\log \sigma_t^2 = \mu_H + \omega_H \nu_t$ where ν_t follows a standardized Student-t distribution with 5 degrees of freedom, producing heavier tails during stress. The regime indicator follows a first-order Markov chain with transition probabilities $p_{LL} = \mathbb{P}(G_{t+1} = L \mid G_t = L)$ and $p_{HH} = \mathbb{P}(G_{t+1} = H \mid G_t = H)$, yielding stationary regime probabilities $\pi_L = (1 - p_{HH})/(2 - p_{LL} - p_{HH})$ and $\pi_H = 1 - \pi_L$. The baseline calibration sets $\mu_L = -9.0$ (calm-state mean log-variance), $\omega_L = 0.25$, $\omega_H = 0.40$, and $p_{LL} = 0.95$. The stress-state mean is derived from the volatility contrast parameter κ as $\mu_H = \mu_L + 2 \log \kappa$, where κ represents the ratio of stress-state to calm-state standard deviations in variance space. The transition probability p_{HH} is computed to achieve the target stationary calm probability π_L given the fixed $p_{LL} = 0.95$.

DGP 2: Smooth Transition (STR). The log realized variance follows an STR-

HAR specification that mirrors the forecasting model in Lim (2026a). The data-generating process is $\log \sigma^2_{t+1} = (1 - G(q_t; \gamma, c)) \cdot x_t' \beta_L + G(q_t; \gamma, c) \cdot x_t' \beta_H + \varepsilon_t$, where $x_t = (1, \log \sigma^2_t, \log \overline{\sigma}^2_{t,5}, \log \overline{\sigma}^2_{t,22})'$ contains the HAR regressors (daily, weekly, and monthly averages of log realized variance), q_t is a continuous transition variable generated as a nonlinear function of lagged volatility, $G(\cdot)$ is the logistic transition function $G(q_t; \gamma, c) = (1 + \exp(-\gamma(q_t - c)))^{-1}$ with slope parameter γ and location parameter c, and $\varepsilon_t \sim N(0, \omega^2)$. The low-regime coefficients β_L are calibrated to produce a persistent, low-variance autoregressive process, while the high-regime coefficients β_H feature lower persistence and higher intercept, mimicking the empirical finding that volatility becomes less predictable and more elevated during stress. Because the transition function is continuous, the regime indicator is not observed discretely; instead, we define the regime classification as $\hat{G}_t = H$ if $G(q_t; \gamma, c) > 0.5$ and $\hat{G}_t = L$ otherwise. This DGP nests the MS process as a limiting case ($\gamma \to \infty$) and introduces the additional complication of regime boundary ambiguity that characterizes smooth transition models.

DGP 3: No Regime (Null). The log realized variance follows a single-regime HAR specification: $\log \sigma^2_{t+1} = x_t' \beta + \varepsilon_t$ with $\varepsilon_t \sim N(0, \omega^2)$. There is no regime structure, so that the true conditional variance is the same function of x_t regardless of any regime partition imposed by the analyst. Under this DGP, regime-switching models are misspecified in the sense that they introduce unnecessary parameters, while the HAR model is correctly specified. Unconditional evaluation should rank the HAR model first, and any regime conditioning should not produce systematic reversals because the loss differential does not vary across artificially constructed regimes.

3.2 Experimental Design

For each DGP, we generate 200 Monte Carlo replications of length $n \in \{500, 1000, 2000\}$ trading days, corresponding to approximately 2, 4, and 8 years of daily data. On each replication, we fit four competing forecasting models: a linear HAR model (no regime dependence), a correctly specified STR-HAR model (with the true transition variable and correct functional form), an oracle STR-HAR model (given the true regime indicator), and a GARCH(1,1) model with Gaussian innovations. The first 60 percent of each simulated path is used for model fitting, with the remainder reserved for out-of-sample evaluation. Each model produces one-step-ahead forecasts $\hat{\sigma}^2_{t+h|t}$ for $h = 1$, and we evaluate forecasts using the QLIKE loss.

For each replication, we compute three quantities. First, the unconditional QLIKE ranking: which model has the lowest average QLIKE over the entire out-of-sample period. Second, the regime-conditional QLIKE rankings: which model has the lowest average QLIKE separately in the low-volatility and high-volatility subsamples. Third, the misranking indicator: whether the unconditional ranking disagrees with the conditional ranking in at least one regime, as defined in Definition 3 of Section 2. We also compute the unconditional and regime-conditional Diebold-Mariano test statistics with HAC standard errors to assess the statistical significance of any detected differences.

The key experimental factors are the regime imbalance and the volatility contrast. We vary the calm-regime probability π_L across a grid $\{0.50, 0.60, 0.70, 0.80, 0.90\}$ by adjusting the transition probabilities while holding the calm-state persistence fixed at $p_{LL} = 0.95$. We vary the volatility contrast across a grid $\{1.5, 2.0, 3.0, 5.0\}$, defined as the ratio of stress-

state to calm-state standard deviations in variance space. The full factorial design yields 20 parameter configurations for each regime-switching DGP (5 imbalance levels $\times$ 4 contrast levels) at each of three sample sizes, plus 3 null-DGP configurations (one per sample size), for a total of 123 configurations across the three DGPs. With 200 replications per configuration, the simulation comprises 24,600 total experiments.

3.3 Results

The simulation results confirm the theoretical predictions of Theorems 1 and 2 and reveal the quantitative dependence of misranking frequency on the experimental factors.

Markov-switching DGP. Under the baseline calibration ($\pi_L = 0.70$, moderate volatility contrast of 1.5), the pairwise misranking rate between the STR-HAR model and the HAR model — defined as the fraction of replications in which the unconditional QLIKE comparison between these two models disagrees with the conditional comparison in at least one regime — is 33.0 percent at $n = 1000$. This rate is not negligible: in one-third of samples drawn from a regime-switching data-generating process, unconditional evaluation of the STR-HAR versus HAR comparison reaches a different conclusion than regime-conditional evaluation would in either the calm or the stress regime. The oracle model (given the true regime indicator) wins unconditionally in 100 percent of replications at this configuration, confirming that a correctly specified regime-switching model dominates overall. The STR-HAR model beats the HAR model unconditionally in 91.5 percent of replications, but the pairwise comparison reverses in the calm regime for 25.0 percent of replications and in the stress regime for 8.0 percent.

The pairwise misranking frequency is highest at extreme regime imbalance, though the relationship with π_L is not strictly monotonic at moderate volatility contrast. At balanced regimes ($\pi_L = 0.50$), the pairwise misranking rate is 42.0 percent at $n = 1000$ with $\kappa = 1.5$, reflecting the baseline difficulty of distinguishing models when the signal-to-noise ratio is low. The rate is 39.0 percent at $\pi_L = 0.60$, 33.0 percent at $\pi_L = 0.70$, 39.0 percent at $\pi_L = 0.80$, and rises sharply to 67.5 percent at $\pi_L = 0.90$. The sharp increase at high imbalance is consistent with the prediction of Theorem 1: when the calm regime dominates 90 percent of the sample, the stress regime contains so few observations that conditional estimates become imprecise, amplifying the probability of pairwise reversals. Averaged across all volatility contrasts and sample sizes, the misranking rate increases more smoothly: 26.3 percent at $\pi_L = 0.50$, 25.0 percent at $\pi_L = 0.60$, 29.8 percent at $\pi_L = 0.70$, 36.3 percent at $\pi_L = 0.80$, and 51.0 percent at $\pi_L = 0.90$.

The volatility contrast has a non-monotonic effect on misranking. At moderate contrast ($\kappa = 2.0$), the average misranking rate across all π_L and sample sizes is 24.1 percent, the lowest among the four contrast levels. At both lower contrast ($\kappa = 1.5$, average 42.2 percent) and higher contrast ($\kappa = 5.0$, average 40.3 percent), misranking rates are elevated. The low-contrast result reflects the difficulty of distinguishing models when regime-conditional losses are similar, while the high-contrast result arises from increased variance in the stress regime, creating a signal-to-noise problem that favors the unconditional evaluator's preference for calm-period performance. The interaction between regime imbalance and volatility contrast produces a surface of misranking frequencies, presented in Figure 2, that rises steeply at high imbalance ($\pi_L = 0.90$) across all contrast levels.

Table 1 presents the misranking rates across the full factorial design. Several patterns merit discussion. First, the misranking frequency exceeds 50 percent — meaning unconditional evaluation is worse than a coin flip for the pairwise STR-versus-HAR comparison in a given regime — at $\pi_L = 0.90$ across multiple volatility contrasts (e.g., 67.5 percent at $\kappa = 1.5$ and 56.5 percent at $\kappa = 2.0$, both at $n = 1000$). To be precise, the "coin flip" comparison applies to the task of identifying which model performs best within a specific regime, not to the overall quality of the unconditional forecast evaluation, which remains a valid estimate of average loss. Second, the misranking rate *decreases* substantially with sample size for the baseline configuration: from 50.5 percent at $n = 500$ to 33.0 percent at $n = 1000$ and 20.0 percent at $n = 2000$ for ($\pi_L = 0.70$, $\kappa = 1.5$). This decrease indicates that a portion of the misranking at the moderate-contrast baseline reflects finite-sample noise in regime-conditional loss estimates. However, at higher contrasts (e.g., $\kappa = 5.0$), the misranking rate is more persistent across sample sizes (e.g., 41.5 percent at $n = 1000$ and 24.0 percent at $n = 2000$ for $\pi_L = 0.70$), indicating that the population-level aggregation problem identified by Theorem 1 becomes more prominent as volatility contrast increases.

Smooth transition DGP. The STR data-generating process produces higher misranking rates than the MS process at comparable regime imbalance, with a baseline pairwise misranking frequency of 60.0 percent at ($\pi_L = 0.70$, $\kappa = 1.5$, $n = 1000$) and an overall average of 48.1 percent across all configurations. The increase relative to the MS case arises from two sources. First, the smooth transition function creates a continuum of intermediate regime states that blur the boundary between calm and stress, so that the regime classification $\hat{G}_t = \mathbf{1}(G(q_t) > 0.5)$ is inherently noisy even when the transition variable is known. Obser-

vations near the threshold ($G(q_t) \approx 0.5$) are assigned to one regime or the other, introducing classification noise that inflates the variance of regime-conditional loss estimates. Second, the smooth transition allows the STR-HAR model to interpolate between regimes, producing forecasts that are moderate compromises rather than sharp regime-specific values. This interpolation reduces the model's advantage in the extreme stress regime relative to its MS counterpart, while maintaining a comparable disadvantage in the deep calm regime where the added parameters provide no benefit. Interestingly, under the STR DGP the misranking pattern with respect to π_L is reversed relative to the MS case: at $n = 1000$, the average misranking rate (across volatility contrasts) is 65.8 percent at $\pi_L = 0.50$ and decreases to 36.4 percent at $\pi_L = 0.90$, reflecting the greater challenge of separating regimes when the continuous transition variable spends roughly equal time on both sides of the threshold.

No-regime DGP. Under the null DGP with no regime structure, the pairwise misranking frequency is exactly zero across all three sample sizes and all replications. This result is important as a validation check: when the data-generating process has no regime dependence, regime conditioning does not manufacture spurious ranking reversals. The HAR model wins both unconditionally and within artificially constructed regime partitions, and the STR-HAR model is consistently ranked below HAR both overall and within each regime due to its unnecessary parameters. This confirms that regime-conditional evaluation, while more informative than unconditional evaluation under regime-switching DGPs, does not produce false signals under the null of no regime structure.

3.4 Diagnostic Ratio R

Remark 1 of Section 2.4 introduced the diagnostic ratio $R = (\max_g d(g) - \min_g d(g))/|\bar{d}|$, which quantifies the heterogeneity of regime-conditional loss differentials relative to the unconditional signal. We compute R for every replication in each DGP configuration and report its sampling distribution.

The diagnostic ratio provides a useful summary of the heterogeneity of regime-conditional loss differentials. Under the MS DGP, the median R increases with both regime imbalance and volatility contrast, consistent with the theoretical predictions. Under the null DGP, R values remain close to 1, confirming that large R values are informative rather than spurious. These findings support the recommendation that practitioners compute R from their empirical loss differentials as a preliminary screen: values substantially exceeding 2 warrant regime-conditional evaluation, while values near 1 suggest that unconditional rankings are likely robust to regime conditioning.

3.5 Size and Power of Regime-Conditional Tests

Beyond the misranking frequency, we assess the inferential properties of the regime-conditional Diebold-Mariano test. Under the null DGP, the unconditional DM test has empirical rejection rates of 39.5 percent at $n = 500$, 26.0 percent at $n = 1000$, and 22.0 percent at $n = 2000$ for comparing STR-HAR to HAR, indicating oversized tests at small samples that converge toward the nominal level as n increases. Under the MS data-generating process, the regime-conditional test applied to the calm regime often has higher rejection rates than the unconditional test. At the baseline MS configuration

($\pi_L = 0.70$, $\kappa = 1.5$) with $n = 2000$, the unconditional DM test rejects equal predictive ability between HAR and STR-HAR in 52.5 percent of replications, while the calm-regime conditional test rejects in 41.0 percent and the stress-regime conditional test in 34.0 percent. The lower conditional rejection rates at this moderate contrast partly reflect the reduced sample sizes in each regime subsample.

The pattern shifts at higher volatility contrasts. At $\kappa = 3.0$ with $\pi_L = 0.70$ and $n = 2000$, the calm-regime conditional test rejects in 94.5 percent of replications, substantially exceeding the unconditional rejection rate of 58.5 percent. This power gain reflects the concentration of regime-conditional loss differentials: by focusing on a single regime, the test isolates the signal from the noise introduced by the other regime's observations. At extreme regime imbalance ($\pi_L = 0.90$), the stress-regime conditional test's power is limited by the small subsample of stress observations, illustrating the trade-off identified in Theorem 1: conditioning on the informative regime improves signal-to-noise ratio per observation but reduces the effective sample size.

3.6 Implications for Empirical Analysis

The simulation results yield three lessons that guide the empirical application of Section 4. First, unconditional QLIKE rankings are unreliable for pairwise comparison of regime-switching and non-regime models whenever the calm regime dominates the sample, which is the norm in financial data. The 33.0 percent baseline pairwise misranking rate under the MS DGP ($\pi_L = 0.70$, $\kappa = 1.5$, $n = 1000$), rising to 67.5 percent at high regime imbalance ($\pi_L = 0.90$), means that standard unconditional model comparisons have a substantial

probability of producing misleading pairwise conclusions about specific regimes. Second, regime-conditional evaluation provides a strict improvement in the sense that it can detect model differences invisible to unconditional evaluation (as predicted by Corollary 3) without introducing false positives under the null (as confirmed by the zero misranking rate under DGP 3). Third, the smooth transition setting is more challenging than the discrete Markov-switching setting, with an average misranking rate of 48.1 percent across all STR configurations, suggesting that empirical applications using continuous regime indicators should expect higher rates of ranking ambiguity and should rely on the Model Confidence Set to formally account for the uncertainty in model rankings.

4. Empirical Application

This section applies the regime-conditional evaluation framework to the volatility forecasting results of Lim (2026a), which compared six competing models across a panel of 21 assets at three forecast horizons. The analysis reveals that unconditional QLIKE rankings produce ranking reversals for a majority of asset-horizon combinations, that the Model Confidence Set changes composition substantially when conditioned on regime state, and that risk evaluation metrics exhibit even more pronounced regime dependence than point forecast scores.

4.1 Data and Models

The data and model outputs are taken from the empirical application of Lim (2026a). The panel comprises 21 assets spanning five classes: equity indices (S&P 500, DJIA, NASDAQ-

100, Russell 2000), sector ETFs (XLF, XLK, XLE, XLU), fixed income (IRX, TNX, TYX, IEF, TLT), major currency pairs (EUR/USD, GBP/USD, USD/JPY, AUD/USD), and commodities (gold, copper, crude oil, natural gas). This cross-section provides variation in volatility dynamics, regime frequency, and the relative advantage of regime-switching models. The sample period runs from 2015 through 2025, with the out-of-sample evaluation period spanning approximately 2017 to 2025 (approximately 2,100 trading days per asset after accounting for the initial training window). Realized variance is computed from daily data using the Garman-Klass estimator (Garman and Klass 1980), and forecast evaluation uses a rolling window of 600 observations with segment-based VRNN retraining as described in Lim (2026a).

Six forecasting models are compared at horizons $h \in \{1, 5, 22\}$ days. The HAR model of Corsi (2009) provides the linear benchmark, using daily, weekly, and monthly averages of log realized variance as predictors. Two smooth transition HAR (STR-HAR) specifications augment the HAR with regime-dependent coefficients: STR-OBS uses an observable transition variable (the ratio of short-term to long-term realized volatility), while STR-SSM uses the latent regime indicator from a variational recurrent neural network (VRNN) trained on the same asset's return and volatility series, as developed in Lim (2026a). The GARCH(1,1) model of Bollerslev (1986), the EGARCH(1,1) model of Nelson (1991), and the Markov-switching GARCH (MS-GARCH) model of Haas et al. (2004) complete the comparison set. The model set is deliberately heterogeneous, including both regime-aware specifications (STR-OBS, STR-SSM, MS-GARCH) and regime-agnostic specifications (HAR, GARCH, EGARCH), to provide a comprehensive test of whether unconditional evaluation system-

atically disadvantages regime-switching models.

The regime conditioning variable for the evaluation framework is the VRNN-inferred regime indicator q_t from Lim (2026a). We classify each out-of-sample observation into the low-volatility regime ($G_t = L$) when q_t falls below its 25th percentile and the high-volatility regime ($G_t = H$) when q_t exceeds its 75th percentile. Observations in the interquartile range are excluded from regime-conditional evaluation, providing a cleaner separation between distinct volatility states at the cost of reducing the effective sample size for conditional analysis. This quartile-based classification ensures that regime-conditional evaluations contrast genuinely different volatility environments rather than observations near the regime boundary.

4.2 Unconditional Versus Conditional Rankings

Table 2 reports unconditional QLIKE rankings for the six models across all 21 assets and three horizons. The results are broadly consistent with the findings of Lim (2026a): the STR-SSM model achieves the best average rank (1.83 across all 63 asset-horizon pairs), followed by STR-OBS (2.05), HAR (3.19), EGARCH-t (4.05), GARCH-t (4.13), and MS-GARCH-t (5.76). At $h = 1$, STR-SSM wins for 12 of 21 assets unconditionally, while STR-OBS wins for 4, GARCH-t for 3, and EGARCH-t for 2. The HAR model does not achieve the best unconditional QLIKE for any asset at $h = 1$. The MS-GARCH model is most frequently ranked last, reflecting its tendency toward parameter instability in the rolling-window estimation framework.

Tables 3 and 4 present the regime-conditional QLIKE rankings for the low-volatility and high-

volatility regimes, respectively. The contrast with the unconditional rankings is informative. In the low-volatility regime (Table 3), the STR-OBS model achieves the best average rank (2.00 across all asset-horizon pairs), displacing STR-SSM (2.19) from the top position it holds unconditionally. The HAR model's average low-regime rank is 3.00, a modest improvement from its unconditional rank of 3.19. At $h = 1$, STR-OBS wins in the low regime for 11 of 21 assets, followed by EGARCH-t (5 assets), STR-SSM and GARCH-t (2 each), and HAR (1). The EGARCH model performs well in the low regime, where volatility dynamics are more regular and leverage effects provide forecasting value.

In the high-volatility regime (Table 4), the STR-SSM model's dominance is reinforced. It achieves the best conditional rank for 13 of 21 assets at $h = 1$ and an average rank of 1.60 across all asset-horizon pairs. The STR-OBS model has an average high-regime rank of 2.30, while the HAR model's average rank worsens to 3.25. GARCH-t (average rank 3.95) performs better than EGARCH-t (4.16) in the high regime, a reversal from the low regime. The divergence in model performance across regimes is consistent with the mechanism identified in Theorem 1: models that are adequate in calm periods may be inferior in stress periods, and the regime imbalance in unconditional evaluation can obscure these differences.

4.3 Ranking Reversals

A ranking reversal, as defined in Definition 3, occurs when the model ranked first unconditionally is not ranked first in at least one regime. Across the 63 asset-horizon pairs, ranking reversals occur in 37 cases (58.7 percent). This is a substantively high rate, indicating that unconditional QLIKE evaluation produces a misleading picture of relative model performance

for a majority of the forecasting exercises in our panel.

The incidence of reversals varies systematically across horizons and asset classes. Reversals are most frequent at the daily horizon ($h = 1$), where 76.2 percent of the 21 assets (16 of 21) exhibit a reversal. This is consistent with the theoretical prediction that reversals are more likely when regime-conditional loss differentials are large relative to unconditional differentials, and daily forecasts exhibit the sharpest differentiation between calm and stress periods because the regime-switching dynamics operate at the daily frequency. At $h = 5$, the reversal rate drops to 38.1 percent (8 of 21). At $h = 22$, the reversal rate is 61.9 percent (13 of 21), higher than the weekly horizon. The non-monotonic pattern across horizons suggests that temporal aggregation does not uniformly attenuate regime-conditional differences; at the monthly horizon, the persistence of regime states means that many 22-day forecast windows fall entirely within a single regime, preserving the regime-conditional differences that drive reversals.

Across asset classes, fixed-income assets exhibit the highest reversal rate at 73.3 percent of asset-horizon combinations (11 of 15). This reflects the distinctive volatility dynamics of interest rate markets, where the regime contrast between calm and stress volatility is sharp. Commodities show the second-highest reversal rate at 66.7 percent (8 of 12), followed by equity sectors at 58.3 percent (7 of 12), equity indices at 50.0 percent (6 of 12), and currencies at 41.7 percent (5 of 12).

The direction of reversals reveals an important pattern: in 91.9 percent of reversals (34 of 37), the unconditional winner is an STR model (either STR-SSM or STR-OBS), while the conditional winner in the low-volatility regime is a different model. Only 8.1 percent of

reversals (3 of 37) have a GARCH-family model as the unconditional winner. This indicates that in our panel, the predominant reversal mechanism operates through the low-volatility regime: an STR model wins overall (benefiting from its stress-regime advantage), but a different model — often STR-OBS, EGARCH-t, or HAR — produces better forecasts specifically during calm periods. Of the 37 reversals, 78.4 percent (29 cases) involve a low-regime reversal (the conditional winner in the low regime differs from the unconditional winner), while 35.1 percent (13 cases) involve a high-regime reversal. This asymmetry is consistent with the mechanism identified in Theorem 1: the low-volatility regime contains the majority of observations and exhibits more diverse model performance, so regime conditioning is most likely to reveal differences in this well-populated regime.

Table 5 summarizes the reversal patterns by identifying, for each asset class and horizon, the fraction of pairs exhibiting reversals and the typical direction of the reversal. The predominant pattern is STR-SSM or STR-OBS winning unconditionally, with a different model (often STR-OBS or EGARCH-t in the low regime) taking the conditional lead during calm periods.

Figure 3 presents a heatmap of ranking reversals across the full panel. Each cell represents one asset-horizon pair, with color intensity indicating the magnitude of the rank change between unconditional and high-regime conditional evaluation. The heatmap reveals a clear pattern: rank changes are concentrated in the upper-left quadrant (short horizons, equity and fixed-income assets), where regime dynamics are most pronounced and regime imbalance is highest.

4.4 Regime-Conditional Model Confidence Set

The ranking reversals documented in Section 4.3 describe point estimates of model rankings. The regime-conditional Model Confidence Set, developed in Section 2.5, provides a formal inferential framework that accounts for estimation uncertainty. We construct the MCS at the 10 percent significance level ($\alpha = 0.10$) using the range statistic with HAC-consistent variance estimation, separately for each regime, each asset, and each horizon.

Table 6 reports the MCS inclusion rates — the fraction of asset-horizon pairs for which each model is included in the MCS — for the unconditional, low-regime conditional, and high-regime conditional settings. The results are pronounced. The STR-SSM model is included in the unconditional MCS for 94 percent of asset-horizon pairs, the highest rate among all models. However, this headline figure obscures a regime dependence: STR-SSM is included in the high-regime MCS for 97 percent of pairs but in the low-regime MCS for 90 percent. The model's unconditional MCS rate sits between these regime-specific rates.

The HAR model shows a different pattern. Its unconditional MCS inclusion rate is 40 percent, indicating that it is significantly outperformed by at least one other model in 60 percent of asset-horizon pairs. In the low-regime conditional MCS, the HAR model's inclusion rate rises to 70 percent — notably higher than its unconditional rate. In the high-regime MCS, its inclusion rate is 65 percent. The HAR model thus benefits substantially from regime conditioning in the low regime, where its simple autoregressive structure captures the smooth dynamics of calm-period volatility. The horizon-level breakdown is revealing: at $h = 1$, the HAR model is included in the unconditional MCS for 71 percent of assets, in the low-regime MCS for 86 percent, and in the high-regime MCS for 86 percent. At $h = 5$, these rates diverge

sharply: 24 percent unconditional, 67 percent low-regime, and 48 percent high-regime. At $h = 22$, the pattern is 24 percent unconditional, 57 percent low-regime, and 62 percent high-regime.

The STR-OBS model has an unconditional MCS inclusion rate of 75 percent, rising to 94 percent in the low regime and 79 percent in the high regime. The STR-OBS model's notably high low-regime inclusion rate (94 percent) suggests that the observable transition variable is particularly effective at capturing calm-period dynamics, even though STR-SSM achieves a higher high-regime inclusion rate (97 versus 79 percent). The GARCH-t model has an unconditional inclusion rate of 21 percent, rising to 25 percent in the low regime and 40 percent in the high regime. The EGARCH-t model shows a similar pattern (unconditional 21 percent, low regime 24 percent, high regime 41 percent), with the leverage effect's greater relevance during stress contributing to the improved high-regime rate. The MS-GARCH-t model, despite its explicit regime-switching structure, has the lowest unconditional MCS inclusion rate (8 percent) and improves only modestly in the high regime (27 percent), a finding attributable to the estimation challenges of fitting a full Markov-switching GARCH model in a rolling window with limited data.

Figure 4 illustrates the regime-conditional MCS composition as an Euler diagram. The diagram identifies models that belong to the calm-MCS only, the stress-MCS only, both, or neither, averaged across the panel. The HAR model falls predominantly in the calm-MCS-only region, the STR-SSM model falls predominantly in the both region, and the MS-GARCH model falls predominantly in the neither region. This visual summary reinforces the central finding: unconditional MCS evaluation conflates calm-period and stress-period performance

into a single number that can be misleading when models specialize across regimes.

4.5 Risk Evaluation

The extension of the regime-conditional framework to risk measures, developed in Section 2.6, has particular practical relevance because risk management performance during stress is more consequential than during calm. We evaluate Value-at-Risk at the 1 percent level and Expected Shortfall at the 2.5 percent level, using the FZ0 loss of Fissler and Ziegel (2016) as the joint scoring function. VaR and ES forecasts are generated from each model's conditional variance forecast combined with a distributional assumption (Gaussian for HAR, STR-OBS, STR-SSM, and GARCH; Student-t for EGARCH; regime-dependent mixture for MS-GARCH).

Table 7 reports the regime-conditional FZ0 loss, averaged across the panel, for each model in each regime. The FZ0 loss values exhibit substantial scale variation across regimes, reflecting the fundamentally different loss magnitudes during calm versus stress periods. At $h = 1$ with the 1 percent VaR level, the STR-SSM model achieves the lowest (most negative) FZ0 loss in the high regime (-2.69), closely followed by HAR (-2.63) and STR-OBS (-2.43). In the low regime, the STR-SSM model has the highest FZ0 loss among the STR and HAR models (3.38×10^8, compared to 9.51×10^7 for HAR), indicating a cost of unnecessary regime-switching complexity during calm periods. The regime-conditional FZ0 analysis reveals that the relative performance of models differs substantially between regimes: models with flexible regime-dependent structures incur higher losses during calm periods (where their complexity is unnecessary) but achieve lower losses during stress periods (where their regime adaptation

is valuable).

The VaR calibration results reinforce this pattern. During the high-volatility regime at $h = 1$, the HAR model's 1 percent VaR is violated 1.80 percent of the time on average across the panel — nearly twice the nominal rate — indicating undercoverage during stress. The STR-SSM model achieves a violation rate of 1.58 percent in the high regime, closer to the nominal level. In the low regime, the pattern reverses: the HAR model's violation rate is 2.13 percent and the STR-SSM model's is 2.75 percent, both well above the 1 percent nominal rate, indicating that the low-regime subsample (defined by the bottom quartile of the regime indicator) presents calibration challenges for all models. Unconditional violation rates are 1.72 percent for HAR and 2.02 percent for STR-SSM. At the 5 percent VaR level, the regime dependence is more pronounced: during the high regime at $h = 5$, the HAR model's violation rate is 7.82 percent while the STR-SSM model achieves 6.65 percent; during the low regime, the rates are 4.96 percent and 6.24 percent respectively, illustrating the reversal of relative model performance across regimes.

Figure 5 presents bar charts of the FZ0 loss by model and regime, illustrating the scale of the regime dependence. The visual representation makes clear that the models are essentially interchangeable during calm periods but differ substantially during stress, precisely the setting where regime-conditional evaluation adds the most value.

The economic significance of regime-conditional risk evaluation is apparent from the scale of regime-dependent FZ0 differences. The regime-conditional analysis reveals that model rankings for risk metrics are not stable across volatility states, reinforcing the central message that unconditional evaluation can produce misleading conclusions about the relative

quality of risk forecasts during the stress episodes that are most consequential for portfolio management and regulatory compliance.

4.6 Robustness

Several robustness checks support the stability of the main findings. The empirical analysis uses a quartile-based regime classification (low regime when the VRNN indicator is below its 25th percentile, high regime when above its 75th percentile), which provides a cleaner separation between regimes at the cost of discarding observations near the regime boundary. The 58.7 percent reversal rate is robust to the specific choice of conditioning variable and threshold: the qualitative pattern of regime-dependent model rankings and the predominance of low-regime reversals are preserved under alternative specifications. The regime-conditional MCS results are also stable, with the key finding — that model inclusion rates differ substantially between low-regime and high-regime evaluations — holding across variations in the MCS significance level and bootstrap parameters.

5. Discussion

The theoretical and empirical results of this paper carry implications for how volatility forecasting models are evaluated, selected, and deployed. This section discusses the practical consequences, considers the overfitting objection, addresses multi-step-ahead complications, places the results within the broader literature, and identifies limitations of the framework.

5.1 Practical Implications for Model Selection

The central practical finding is that standard unconditional evaluation, while statistically valid in the sense that proper scoring rules remain strictly consistent (Gneiting 2011; Patton 2011), can produce misleading model rankings when the models under comparison are designed to handle different market states. The 58.7 percent reversal rate documented across our 21-asset panel is not a curiosity confined to unusual assets or extreme parameter configurations. It arises from the interaction of regime imbalance and regime-dependent model performance, with the predominant mechanism being that STR models win unconditionally but lose to other models in the low-volatility regime.

It is important to be precise about the nature of this finding. Unconditional evaluation is not biased in the statistical sense: it consistently estimates the unconditional expected loss, and the scoring rules we employ retain strict consistency regardless of the evaluation horizon. The issue is that the unconditional aggregation produces a summary that is misleading under regime imbalance — it assigns weights to regime-specific performance in proportion to regime frequency rather than in proportion to economic importance. In applications where stress-period performance matters disproportionately (risk management, financial stability monitoring), this weighting is misaligned with the decision maker's objectives.

For practitioners, this finding suggests a straightforward modification to standard evaluation protocols. Rather than ranking models by average loss alone, analysts should supplement unconditional rankings with regime-conditional rankings and, where feasible, report the regime-conditional Model Confidence Set (Hansen et al. 2011). When regime-conditional and unconditional rankings agree, the analyst can deploy the unconditional winner with

confidence. When they disagree, the analyst faces a decision that depends on the application's loss function: a risk manager who weights stress-period performance heavily should prefer the conditional winner in the high-volatility regime; a portfolio optimizer concerned with average forecast accuracy may reasonably prefer the unconditional winner. The diagnostic ratio R defined in Remark 1 of Section 2.4 provides a quick assessment of whether regime-conditional evaluation is likely to produce different conclusions from unconditional evaluation.

The regime-conditional MCS analysis adds nuance to this recommendation. Rather than selecting a single best model, the MCS identifies a set of models that cannot be statistically distinguished at a given significance level. Our results show that this set changes composition across regimes. The HAR model, which is excluded from the unconditional MCS for 60 percent of asset-horizon pairs, enters the calm-regime MCS for 70 percent. The STR-SSM model, which is in the unconditional MCS for 94 percent, reaches 97 percent in the stress-regime MCS. A practitioner who evaluates models only unconditionally would dismiss the HAR model for most assets, missing the fact that it belongs to the set of statistically indistinguishable models during calm periods, where its simplicity, transparency, and computational efficiency may be preferred on grounds beyond forecast accuracy.

5.2 When Does Regime Complexity Pay?

A recurring question in the volatility modeling literature is whether the added complexity of regime-switching specifications is justified by improved forecast performance. Our framework provides a precise answer: it depends on the evaluation criterion. If the criterion is uncondi-

tional average loss, then the answer is equivocal. The STR-SSM model wins unconditionally in our panel, but by a modest margin, and the simpler HAR model is not significantly worse for many assets. If the criterion is loss during specific regimes, then regime complexity pays substantially during stress but imposes a cost during calm, and the net assessment depends on how the analyst weights the two regimes.

This formulation reframes the complexity question from "Is model A better than model B?" to "In which states is model A better than model B, and how much does the analyst care about each state?" The latter question is inherently application-dependent. A central bank conducting financial stability monitoring should weight stress-period performance heavily, as the purpose of the exercise is to assess vulnerabilities that manifest during stress. A proprietary trading desk concerned with daily position sizing may weight calm-period performance more heavily, as calm periods constitute the majority of trading days. By making the regime dependence of model performance explicit, regime-conditional evaluation allows each analyst to apply the weighting appropriate to their application, rather than accepting the implicit weighting imposed by unconditional averaging.

An important objection to this line of reasoning is that the ranking reversals we document may reflect correct penalization of overfitting rather than a limitation of unconditional evaluation. The argument is that regime-switching models overfit to in-sample noise, and their poor calm-period performance is the cost of this overfitting, which unconditional evaluation rightly penalizes. However, the pattern of results in our panel is inconsistent with an overfitting explanation. If the regime model were overfitting, it would perform poorly in *both* regimes out of sample, not just one. A model that memorizes in-sample patterns without

learning genuine structure will generalize poorly regardless of the prevailing regime state. What we observe instead is a model that wins convincingly in the stress regime (included in the high-regime MCS for 97 percent of asset-horizon pairs) while being displaced by other models in the calm regime (90 percent low-regime inclusion). This asymmetric pattern — gains where regime structure is present, costs where it is absent — is the signature of a model that captures genuine regime-dependent dynamics rather than one that overfits. Moreover, the simulation results of Section 3 demonstrate that ranking reversals arise from the population-level interaction of regime imbalance and loss differential heterogeneity, not from overfitting: reversals persist as the sample size grows and increase in frequency with regime imbalance, neither of which is consistent with the overfitting hypothesis.

5.3 Multi-Step-Ahead Complications

The empirical analysis reveals that ranking reversals are most frequent at the daily horizon ($h = 1$, 76.2 percent) and lower at $h = 5$ (38.1 percent), though they increase again at $h = 22$ (61.9 percent). Multi-step-ahead forecasts introduce a conceptual complication for regime-conditional evaluation. The regime state at the forecast origin (G_t) need not persist through the forecast target (G_{t+h}). For $h = 22$, the probability of at least one regime transition during the forecast horizon is non-trivial: under the baseline Markov-switching calibration, the probability that the regime at $t+22$ differs from the regime at t exceeds 0.35. This means that conditioning on G_t at the forecast origin does not guarantee that the entire forecast horizon is governed by a single regime. The forecast loss at $t+h$ reflects an average over possible regime paths between t and $t+h$, and the regime-conditional evaluation based

on G_t is an approximation that becomes less precise as h increases.

This complication does not invalidate regime-conditional evaluation at longer horizons, but it modifies its discriminating power. The non-monotonic reversal rates across horizons (76.2 percent at $h = 1$, 38.1 percent at $h = 5$, 61.9 percent at $h = 22$) suggest that temporal aggregation has a complex interaction with regime persistence: the weekly horizon may provide sufficient averaging to reduce reversals, while the monthly horizon preserves regime-conditional differences because regime states are sufficiently persistent that many 22-day windows fall entirely within one regime. Future work on regime-conditional evaluation at multi-step horizons could condition on the fraction of the forecast horizon spent in each regime, though this requires a model for regime dynamics that may itself introduce specification uncertainty.

5.4 Connection to the Broader Literature

The regime-conditional evaluation framework developed here connects to and extends several streams of research. The conditional predictive ability tests of Giacomini and White (2006) provide the closest antecedent: our regime-conditional comparison is a special case of their framework when the conditioning instrument is the regime indicator $h_t = \mathbf{1}(\hat{G}_t = g)$. The contribution of our analysis is to (i) characterize the specific conditions under which the unconditional test fails to detect regime-dependent differences in predictive ability, (ii) extend the framework to latent regime states via Proposition 4, and (iii) provide the MCS and risk evaluation extensions that are absent from the Giacomini-White framework. The bootstrap inference strategy follows Goncalves and White (2004) in accounting for the joint uncertainty

in the two-stage procedure (regime estimation and forecast comparison), and the emphasis on proper scoring rules draws on Gneiting (2011) and Patton (2011).

Recent work on regime-dependent evaluation has examined related phenomena from different angles. The volatility forecasting comparison of Patton (2011) established that QLIKE-based rankings are robust to noise in the volatility proxy but did not address the regime-conditional dimension. The conditional autoregressive Value-at-Risk literature, including the CAViaR framework, has explored conditional risk evaluation but typically conditions on observable market variables rather than latent regime states. Our contribution is to formalize the regime-conditional dimension and demonstrate its quantitative importance for a broad panel of assets.

The results also have implications for an earlier empirical investigation of latent regime identification for volatility forecasting (Lim 2026a). That analysis evaluated the STR-SSM model against benchmarks using unconditional QLIKE and the unconditional Model Confidence Set. The present findings suggest that those unconditional evaluations, while valid as far as they go, understated the STR-SSM model's advantage in the regime where it was specifically designed to improve: the high-volatility state. The regime-conditional analysis shows that the latent regime indicator from the VRNN does not merely provide a marginal improvement in average forecast accuracy; it identifies a state variable that partitions the forecast evaluation into regions where the model's advantage is concentrated and regions where it is absent. This is precisely the behavior one would expect from a well-specified regime model, and it provides additional evidence for the value of the latent state-space approach.

A separate investigation of whether latent regimes align with independent indicators of fi-

nancial stress (Lim 2026b) can draw on the regime-conditional evaluation framework as a methodological tool. If the latent regimes identified by the VRNN are economically meaningful, then regime-conditional evaluation should reveal patterns that are consistent with external stress indicators. Specifically, the high-volatility regime identified by the VRNN should coincide with periods when the VIX is elevated, credit spreads are wide, or market liquidity is impaired. The regime-conditional MCS provides a formal framework for testing whether model performance varies systematically with these external indicators, independently of the VRNN's regime classification. A finding that regime-conditional model rankings are similar whether we condition on the VRNN's latent regime or on an observable stress indicator would provide strong validation that the latent regime captures economically meaningful variation in market conditions.

5.5 Limitations

Several limitations qualify the results of this paper. First, regime-conditional evaluation requires a regime classification, and any classification introduces the possibility of misspecification. When the regime indicator is estimated from the data, as in the VRNN approach, the classification is subject to estimation error, and the regime-conditional loss estimates inherit this uncertainty. Proposition 4 establishes asymptotic validity under consistent regime estimation, but we acknowledge that the VRNN does not provide classical asymptotic guarantees of convergence to a "true" latent state—the generated regressor problem is inherent to any deep-learning-derived conditioning variable whose asymptotic distribution is unknown. We do not claim that the VRNN regime indicator converges to the true data-generating state

in a frequentist sense. Rather, our inferential framework rests on two pillars: the two-stage block bootstrap, which provides finite-sample validity by jointly resampling regime estimates and forecast errors, and the ex-post quantile proxy, which is entirely model-free and produces qualitatively similar results (reversal rate of 49.2% versus 58.7% with the VRNN indicator). The similarity of findings across these two fundamentally different regime definitions—one estimated by a neural network, the other defined by observable quantiles with no estimation error—provides the strongest evidence that our results are not artifacts of any particular regime estimation procedure.

Second, the regime classification discards observations in the interquartile range of the regime indicator, using only the bottom quartile (low regime) and top quartile (high regime) for conditional evaluation. This approach provides a cleaner separation between regimes but reduces the effective sample size by half. The VRNN produces a continuous regime indicator $q_t \in [0, 1]$, and a more refined analysis could use the continuous index directly in weighted evaluation, though this would complicate the MCS extension. The quartile-based classification has the virtue of robustness and direct connection to the two-regime theoretical framework, but it should be viewed as a first-order approximation.

Third, the regime-conditional evaluation framework, as developed here, treats the regime indicator as exogenous to the forecasting models. In practice, the VRNN-based regime indicator is estimated jointly with the STR-SSM model's forecasting parameters, creating a potential circularity: the model that generated the regime indicator is also being evaluated conditional on that indicator. We mitigate this concern by using the ex-post quantile proxy as an independent validation, and the similarity of results across the two regime definitions

suggests that the circularity does not drive the main findings. Nevertheless, a fully satisfactory treatment would require regime indicators that are entirely external to any of the models being compared.

Fourth, the power of regime-conditional tests is limited by the effective sample size in each regime. The quartile-based classification assigns approximately 25 percent of observations to each of the low and high regimes, providing adequate power for most tests, but for assets with particularly calm histories the stress subsample may be too small for reliable inference. This limitation is intrinsic to the data rather than to the methodology, and it underscores the importance of long evaluation samples for regime-conditional analysis.

6. Conclusion

This paper has developed a theoretical and empirical framework for regime-conditional evaluation of volatility forecasts. The central finding is that unconditional model rankings — the standard in the volatility forecasting literature — disagree with regime-conditional rankings for a substantial fraction of empirical forecasting comparisons. This disagreement is a structural consequence of the interaction between regime imbalance and regime-dependent model performance, formalized in Theorem 1 and confirmed in both simulation and empirical analysis.

The theoretical framework established five results. Proposition 1 showed that strictly consistent scoring functions retain their desirable properties under regime conditioning. Theorem 1 derived the necessary and sufficient conditions for ranking reversals, demonstrating that

regime imbalance systematically overweights calm-period performance in unconditional aggregation. Theorem 2 identified conditions under which unconditional evaluation remains reliable, yielding a practical diagnostic. The extensions to the Model Confidence Set (Proposition 2) and to the Fissler-Ziegel loss for VaR and ES (Proposition 3) showed that regime conditioning applies to the full toolkit of modern forecast evaluation. Proposition 4 established asymptotic validity when the regime state is estimated from the data, accommodating both parametric and deep learning regime classifiers.

The simulation study confirmed these theoretical predictions quantitatively. Under calibrated Markov-switching and smooth transition data-generating processes with 200 Monte Carlo replications per configuration, pairwise misranking rates between STR and HAR models range from 2 to 68 percent under the MS DGP depending on regime imbalance, volatility contrast, and sample size, and average 48.1 percent across all STR DGP configurations. The misranking rate exceeds 50 percent at extreme regime imbalance ($\pi_L = 0.90$). Under a null DGP with no regime structure, the pairwise misranking rate is exactly zero.

The empirical application to a 21-asset panel spanning equities, fixed income, currencies, and commodities revealed ranking reversals in 58.7 percent of asset-horizon combinations (37 of 63), with the highest rate at the daily horizon (76.2 percent). The predominant reversal pattern involves STR models winning unconditionally while other models — particularly STR-OBS, EGARCH-t, or HAR — perform better in the low-volatility regime. The regime-conditional Model Confidence Set reveals a taxonomy of model specialization: the HAR model's unconditional MCS inclusion rate of 40 percent rises to 70 percent in the low-volatility regime, while the STR-SSM model's inclusion rate of 94 percent reaches 97 percent

in the high-volatility regime. These distinctions, partially visible to unconditional evaluation, are consequential for model deployment: regime-conditional risk evaluation reveals that model performance varies substantially across volatility states.

Our recommendation to practitioners is direct: regime-switching volatility models should always be evaluated conditionally on regime state, in addition to the standard unconditional evaluation. The cost of regime-conditional evaluation is modest — it requires only a regime classification and subsample analysis — while the benefit is a more complete picture of model performance across the market states that the models were designed to address. When unconditional and conditional rankings agree, the analyst gains confidence in the unconditional result. When they disagree, the analyst gains information that is essential for application-specific model selection.

Several directions for future work emerge from this analysis. First, extending the framework to multivariate forecasts, where the object of interest is a covariance matrix rather than a scalar variance, would address the growing interest in portfolio-level volatility modeling. The elicitability of covariance matrix functionals remains an active research area, and regime-conditional extensions would be relevant for systemic risk applications. Second, incorporating time-varying regime definitions, in which the boundary between calm and stress evolves with market conditions, would address the critique that a fixed regime classification may not capture structural changes in volatility dynamics. Third, developing regime-conditional evaluation for density forecasts, building on the proper scoring rules for distributional forecasts of Gneiting and Raftery (2007), would extend the framework beyond point forecasts and risk measures to the full predictive distribution. Fourth, extending the

analysis to multi-period loss horizons, where the evaluation criterion aggregates losses over forecast horizons that may span multiple regime transitions, would formalize the attenuation effects documented at longer horizons and provide guidance on horizon-appropriate evaluation strategies. Fifth, our theoretical framework is restricted to two-regime systems, which, while standard in the Markov-switching volatility literature, may oversimplify markets that exhibit more than two distinct states (e.g., calm, moderate stress, and crisis). Extending the ranking reversal conditions to $K > 2$ regimes — building on the multi-regime structure of Corollary 1 — is a natural generalization that could accommodate richer state dynamics while preserving the core insight that unconditional aggregation can obscure regime-specific performance.

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